

An Experimental Platform for Multi-Ion Quantum Control

Scalable Error Characterisation
and Hybrid Resources



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Apparatus

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After 70 years of ion confinement development (since the quadrupole mass spectrometer in the 1950s [1]), and the myriad use-cases from time standards, to analytic chemistry, to quantum computing, there are naturally many systems and theses apparatus chapters in existence today. As such, we structure this chapter in the following way: Following on from the standard requirements of an ion trap

system, provided at the beginning of chapter ??, we discuss the unique motivation and design specifications of our system; We then present an abstract schematic of the system in its current form, highlighting the constituent subsystems, and the optical and electrical signal paths; The remaining sections of this chapter contain technical details, perhaps more relevant to the practicing experimental student, and the hardware enthusiastic.

1.1 Specification

Ion-trap experiments are exceptionally versatile in their application, however in the vast search space of quantum-information applications, we want a system particularly tailored for three avenues of study:

1. Error channels on multi-qubit chains.
2. Continuous variable quantum computing and simulation utilising the spin and motional degrees of freedom of the confined ions.
3. Fast two-qubit entangling gates via phase stabilised optical standing waves.

These latter two research directions were explored within our group on a previous apparatus, leading to state-of-the-art two-qubit entanglement rates on the ion trap platform [2, 3], creation of highly non-classical motional states [4, 5], and a simulation of Z2 gauge theories encoded via the spin and motional degrees of freedom of a two-ion chain [6, 7]. However, hard limitations of both scaling these demonstrations to longer ion chains, and of interaction strength due to available optical intensity, were often restrictive. To solve both these issues, a new apparatus has been designed, constructed, and characterised, motivated by these applications. Colloquially the system is named “*FastGates*”, owing to its experimental objectives, and we shall refer to it as such throughout this chapter.

These objectives give the following specific control requirements:

- Storage of medium length (5-10) qubit registers, section 1.2.

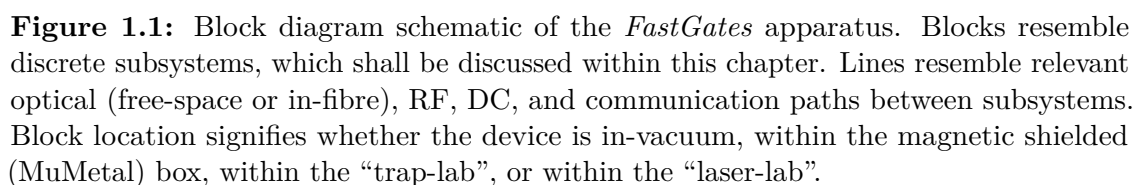
- Individual internal state control of each qubit, section 1.6.5.
- Stable motional modes for storing quantum information, section 1.2.1.
- Dual high numerical aperture (NA) optical access for high-intensity standing waves, section 1.4.

As the lifetime of such an apparatus extends past a single PhD generation, this thesis serves as a point-in-time description of the system, and where applicable, the planned future extensions will be highlighted as such. An abstract schematic of the current configuration is shown in figure 1.1 with blocks resembling either electronic, mechanical, or optical subsystems.

1.2 The Ion Trap

Recently, the micro-fabricated surface style linear Paul trap has gained popularity due to the maturity of chip fabrication technologies [9] and the potential route to scalability this offers. In the surface trap, the 3D radial and axial electrodes of a “macro” trap are effectively projected onto a 2D surface. The stable confining point of such a trap is typically designed to be on the order of $50 - 100 \mu\text{m}$ above the chip surface. The maturity of chip fabrication has enabled the manufacturing of complex multi-zone devices with many DC electrodes, amenable to ion shuttling [10, 11], and hence furthering progress towards Quantum CCD type architectures [12]. However, this surface style geometry typically comes with two costs: the depth of the trapping potential is often greatly reduced due to anharmonicity [13], and the close proximity of the surface to the ion can be a large contributor to motional heating rates [14–17].

Our trap, provided by the *National Physical Laboratory* in the UK, brings together the advantages of chip fabrication as well as the low heating rates and high trapping depths of a macro 3D style trap through utilising a micro-fabricated 3D trap geometry with greater ion-surface distances than typical planar traps. Details on its design and characterisations can be found in Ref. [8, 18, 19]. Figure 1.2 shows



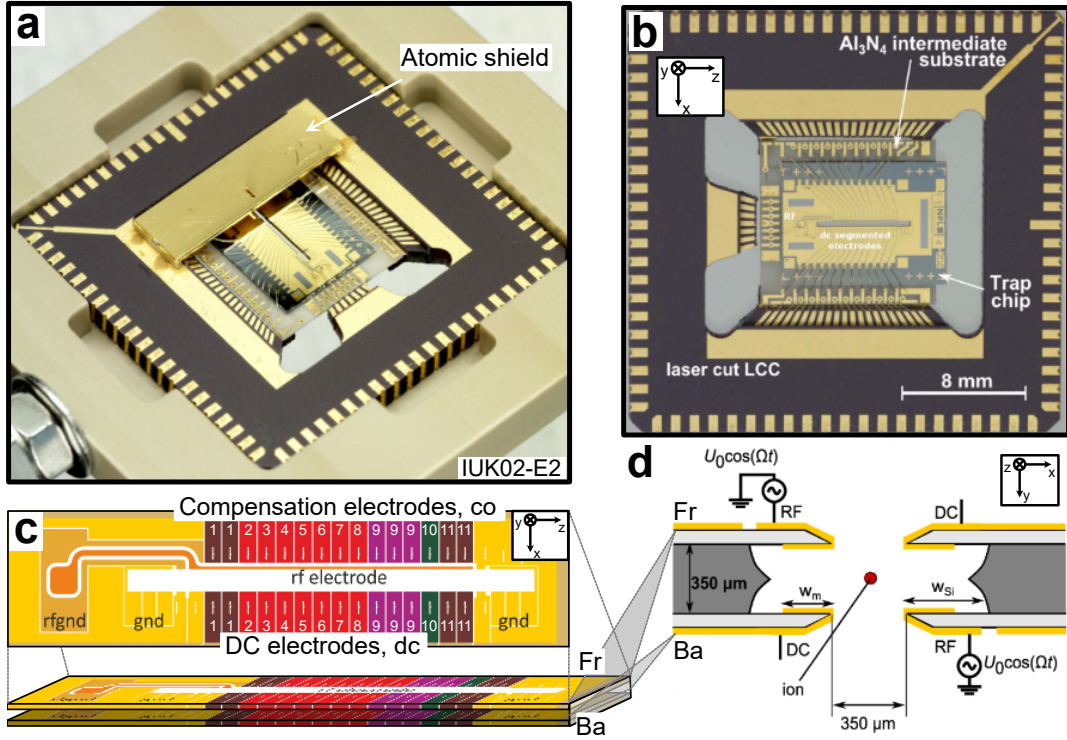


Figure 1.2: Schematics of the *NPL* trap chip used in this experiment. **a)** A photograph of the trap chip used in the *FastGates* experiment before being placed under vacuum. The top “atomic shield” can be seen here. **b)** A front view of the trap chip. Axial confinement is provided by a subset of the segmented DC electrodes. Radial confinement is provided by the RF rails. The trap chip and leadless chip carrier (LCC) are highlighted. **c)** A schematic of the electrode geometries and labelling for the front-face of the trap chip. A similar arrangement exists for the back-face however with the RF rail on the diagonally opposite electrode. **d)** A slice view of the trap. The red point here represents the ions, the gold are the RF and DC electrodes, and the grey is the silica substrate. It can be seen that the ion is $350 \times \sqrt{2}/2 \simeq 250 \mu\text{m}$ away from the nearest electrode. The numerical aperture of the ion plane due to the electrodes is ~ 0.71 . Figures b) and d) adapted from Ref. [8], other panels adapted from private communication with the *NPL* team.

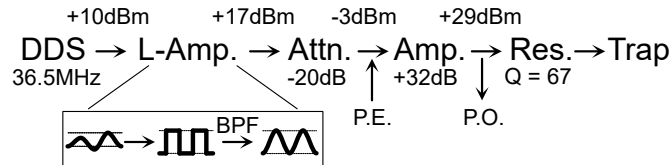
the electrode geometry of the trap and the relevant length scales. The trap package consists of an outer leadless chip carrier (LCC) with outer pads routing DC, RF, and ground connections to an intermediate Al_3N_4 substrate. The connection to the front and back surfaces of the trap chip is made through vias and wire bonds to the intermediate substrate [20]. The trap chip is a monolithic double-sided silicon device with gold plated electrodes on the front (Fr) and back (Ba) surfaces. These two layers are shown schematically in the bottom panel of (c). The electrode schematic for the front surface is shown, and the electrode indexing convention

is highlighted. Electrodes recessed from the ion due to the RF rail are named “compensation”-electrodes (co), while electrodes on the opposite side provide the strong DC potential required for axial confinement¹. We typically trap ions centered on electrode dc5. Panel (d) shows a slice view of the trapping region, highlighting the ion position between the front and back surface electrodes. The dual RF rails and DC electrodes are shown in gold, and the undercut silicon bulk is highlighted in dark grey.

The next two subsections describe the source of RF and DC voltages to the trap electrodes. The physical mounting of the trap in the vacuum system is discussed in section 1.4, and the optical access to the trap is discussed in section 1.4.

1.2.1 Trap RF Chain

To utilise radial motional modes for low error quantum gates, we require the radio frequency (RF) field that produces the desired trapping pseudopotential to be both frequency and amplitude stable. Furthermore, as the natural time scale for two-qubit gate durations is $\tau = 2\pi/\omega_m$ (see section ??), where ω_m is the motional mode angular frequency, then greater motional mode frequency is desired for fast entanglement. From equation ??, it can be seen that radial motional mode frequency is proportional to RF voltage, and inversely proportional to the frequency. The *NPL* trap used in the *FastGates* apparatus is rated for a max peak-to-peak voltage of 400 Vpp on the RF electrodes. To target high motional mode frequencies, saturating this maximum voltage is desirable, however, for safe operation, typically the trap is provided a maximum RF voltage $V_{RF} = 360$ Vpp. The RF chain which provides this signal is schematically shown as:



¹Note: Compensation electrodes co1 and co11 on both front and back surfaces are grounded on the trap package, and so are not available for use. There is a continuity fault on front dc11 and so this electrode is also not available for use.

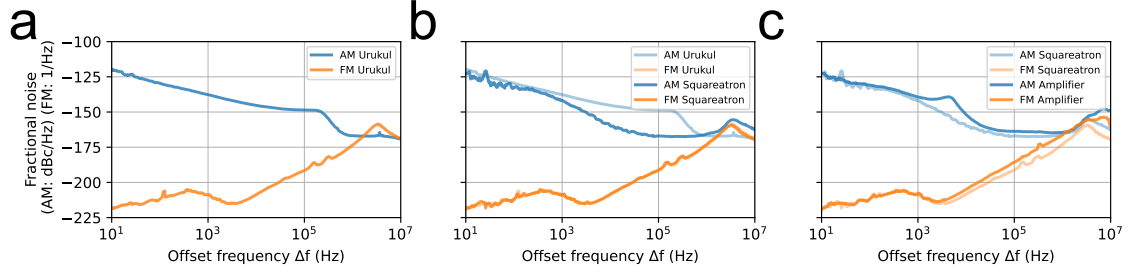


Figure 1.3: Characterisation of the RF chain. **a)** SSB spectra for AM and PM, given as fractional noise, measured after the *Urukul*. **b)** Measured after the *Squareatron*. **c)** Measured after the amplifier.

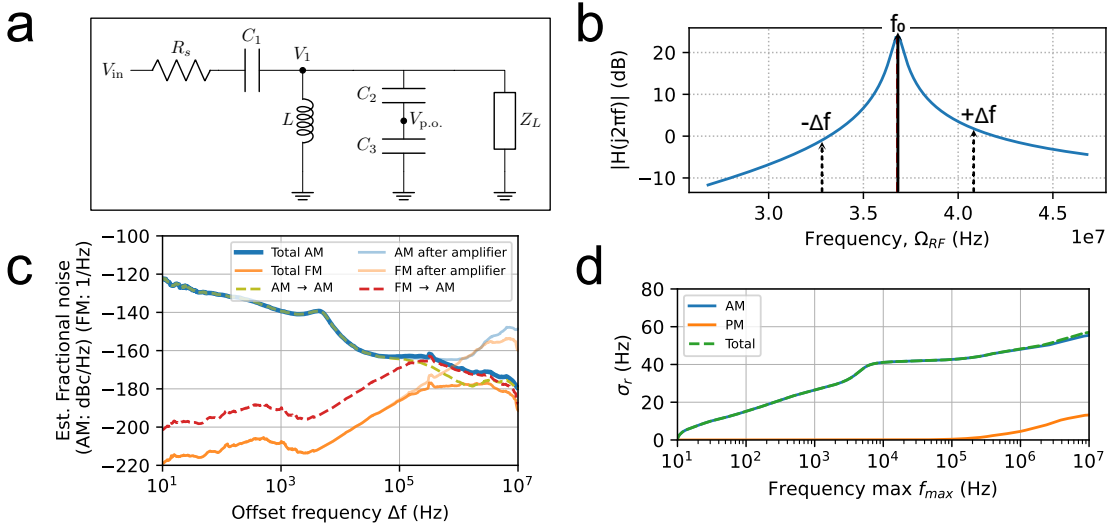


Figure 1.4: Expected SSB after the resonator, and estimated mode frequency noise. **a)** Resonator circuit diagram. **b)** Resonator lineshape, given by magnitude of the transfer function $|H(f)|$. **c)** Expected AM and FM fractional noise spectra after the resonator. **d)** Integrated RMS noise of radial motional mode frequency due to the RF chain.

and embedded within the schematic 1.1, beginning with the *Trap FPGA*. Each element of the chain is designed to provide a low-noise, high-voltage RF signal to the trap electrodes, and is discussed in turn below.

The frequency source is a direct-digital-synthesiser (DDS), named *Urukul* [21], part of the *Artiq Sinara* [22] hardware ecosystem. *Urukul* is used extensively in our lab for driving acousto-optic modulators (AOMs) and here used as the source drive of our trap RF. The *Urukul* is operated at maximum output power, +10 dBm, and with a frequency ~ 35 MHz (set by resonator, discussed below). As can be seen in

equation ??, amplitude and frequency instability will directly lead to radial mode instability. Expanding for small amplitude, dV_{RF} , and drive frequency, $d\Omega_{RF}$, errors we find the motional mode frequency, ω'_r , is related to the ideal frequency, ω_r , by

$$\omega'_r = \omega_r(1 + dV_{RF}/V_{RF} - d\Omega_{RF}/\Omega_{RF}), \quad (1.1)$$

where we can see both errors affect the motional mode frequency linearly. The instability of the oscillator can limit motional mode coherence time (see section 2.2.4 for characterisation with the ion). We measure noise originating from both amplitude modulation (AM) and frequency modulation (FM) across the RF chain using a phase noise analyser². This was done to verify low-noise operation of the individual devices, and to estimate expected motional mode errors due to the inherent noise on the RF chain.

We quote FM as the fractional frequency stability power spectral density (PSD),

$$S_y(f) = \frac{f^2}{f_0^2} S_\phi(f), \quad (1.2)$$

where $S_\phi(f)$ is the phase noise PSD (which is the Fourier transform of the Autocorrelation function) f is the offset frequency, and f_0 is the carrier frequency. $S_y(f)$ has units of $\left[\frac{1}{\text{Hz}}\right]$. AM noise is quoted as the single sideband (SSB) power, S_α , in a 1 Hz bandwidth at an offset frequency, f , from the carrier, normalised to the total power of the carrier, and is here given in units of $\left[\frac{\text{dBc}}{\text{Hz}}\right]$.

Figure 1.3, panel (a) displays the *Urukul* AM and FM spectra. The *Urukul* is phase referenced to the master FPGA board (*Kasli*), which is in turn phase locked to an external 10 MHz atomic clock³. It can be seen that AM noise dominates up ~ 1 MHz, suggesting that this will be the limiting error source for motional mode stability. The AM noise may be reduced by using a non-linear voltage clamp. We use an “ultra-low noise limiting amplifier” named *Squareatron* developed previously in the group, and further refined by *Oxford Ionics*. *Squareatron* takes an input RF signal and converts to a stable amplitude square-wave output which is subsequently bandpassed around the desired RF frequency to recover the sinusoidal signal, but

²Rhode and Schwartz FSWP PNA

³**ToDo: Abaqus atomic clock PN**

with significantly reduced AM noise. The AM and FM after the *Squareatron* is shown in panel (b). It can be seen that between 1 kHz and 1 MHz, AM is considerably suppressed. There appears to be an increase in AM at higher frequencies **ToDo: explore this further**. The *Squareatron* has a 17 dBm output power, which is attenuated by -20 dB before an amplification stage to reach the desired power for the trap. Highlighted in the above chain schematic is the option to input a parametric excitation signal (P.E.), which we shall discuss further in section ???. For the following measurements, the P.E. branch was removed. AM and FM introduced by the low-noise amplifier⁴, can be seen in panel (c). Most notably an increase in AM is seen between 1 kHz and 10 kHz.

To impedance match the $50\ \Omega$ line from the amplifier to the small capacitance of the trap electrodes and vacuum feedthrough, a resonant matching circuit is used. This “resonator” has three main effects on our system:

1. The source $50\ \Omega$ is matched to the (almost) purely capacitive load.
2. The voltage step up provides the necessary amplitude for high motional mode frequencies.
3. The input RF is bandpassed around the desired trapping frequency.

The latter two points are practically useful, especially the bandpass effect of the resonator attenuating noise on the RF line, however, changes to the resonators Q factor and central frequency due to thermal and mechanical noise can be a significant source of low frequency motional mode instability. Opting for good passive stability, a PCB based design⁵ with good thermal management was seen as preferable to large helical resonators, which are often used in ion trap experiments for their high Q factors. Aiming for a moderate Q factor of ~ 70 allows for a good voltage step up, and some bandpass filtering, without being overly sensitive to thermal and mechanical noise. The circuit diagram of the constructed resonator⁶

⁴Mini-Circuits ZHL-1-2W-S+

⁵This style PCB resonator has been used around the Oxford Ion Trapping Group for some time, and is based originally on a design by Tom Harty

⁶Credit to Emily Hirsch for construction of the recent resonator.

is shown in figure 1.4 (a). The inductor, $L \sim 0.8 \mu\text{H}$, is hand-wound copper wire with a toroidal iron powder core⁷, and the capacitors are low tempco C0G rated ceramic, $C = 5.4 \text{ pF}$. The vacuum can feedthrough and trap were measured to present an $\sim 18 \text{ pF}$ load. The trap itself presents a $C_T = 4 \text{ pF}$ [20], the cable between the resonator and the vacuum feedthrough contributes $\sim 4 \text{ pF}$, and the remaining presented capacitance is expected to originate from the feedthrough and in-vacuum RF line. In vacuum, the RF line is a *Kapton* coated 0.4 mm diameter single strand copper wire, which connects to an in-vacuum distribution PCB. The RF connection to the trap LCC is made by *fuzz button* vias.

Measuring the motional mode frequency of a trapped ion while varying the trap RF frequency (see section ??), yields a resonator quality factor $Q = 57.88(6)$, and central frequency $34.659(1) \text{ MHz}$. The effect of the resonator on the AM and FM noise can be calculated through the transfer function, $H(f)$, of the resonator circuit. Capacitors ($C2$ and $C3$) are used to provide a pick-off (p.o.) voltage which can be monitored, and if required, used in a closed feedback loop. We however have not attempted this as of yet, and shall ignore the effect of the pick-off capacitors for this analysis. The transfer function of the simplified circuit is then given by,

$$H(f) = \frac{V_{out}}{V_{in}} = \frac{Z_p}{Z_s + Z_p}, \quad (1.3)$$

where

$$\begin{aligned} Z_p &= \left(\frac{1}{j2\pi fL} + \frac{1}{Z_L} \right)^{-1}, \\ Z_s &= R_s + \frac{1}{j2\pi fC_1}. \end{aligned} \quad (1.4)$$

Given the approximate capacitance and inductance values of the resonator and trap, the analytic $|H(f)|$ is plotted in figure 1.4 panel (b), and shows acceptable agreement with the measured quality factor and central frequency (here $Q = 69$ and $f_0 = 36.8 \text{ MHz}$) with the discrepancy of quality factor expected to originate from excess resistance in the inductor not accounted for in the model. Final AM and FM spectra are calculated through considering both upper ($+\Delta f$) and lower

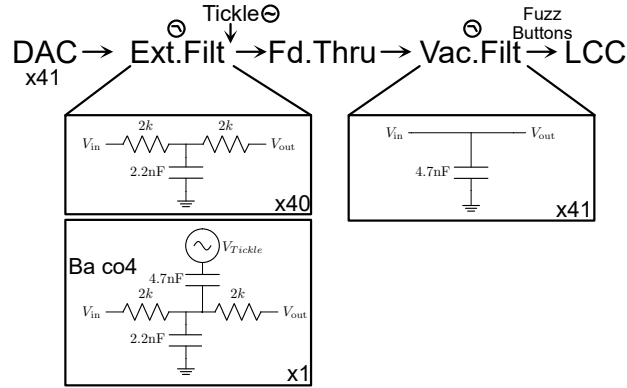
⁷Micrometals T80-6

sidebands ($-\Delta f$), as shown in panel (b), following reference [23]. We find that for both AM $\rightarrow$ AM and FM $\rightarrow$ FM (where “ $\rightarrow$ ” represents the transform from input noise type to output noise type) the input PSD are low pass filtered with a cut-off frequency of ~ 0.25 MHz. We also consider the cross term, FM $\rightarrow$ AM, which here we expect to be non-negligible at higher offset frequencies due to the asymmetry of the resonator lineshape. Panel (c) shows the expected fractional noise after the resonator, with total FM noise (orange line), total AM noise (blue line), and contributions from AM $\rightarrow$ AM (dashed green), FM $\rightarrow$ AM (dashed red). Integrating the AM and FM fractional noise spectra between some chosen range of offset frequencies, f_{min} and f_{max} , gives the expected fractional RMS noise on the radial mode frequency of the ions. Assuming a mode frequency of $w_r = 3$ MHz the RMS, σ_r , is shown in figure 1.4 panel (d) as a function of f_{max} , with $f_{min} = 10$ Hz. It can be seen that the expected noise is dominated by the AM contribution, and in the frequency range considered, corresponds to a mode frequency instability of $\sigma_r = 60$ Hz. This analysis excludes any noise less than 10 Hz, and environmental noise coupled in by the resonator, which may be a significant contribution at low frequencies due to thermal and mechanical noise. The motional modes will also be subject to noise from other coupling mechanisms, such as voltage noise on the DC electrodes, or fluctuating stray electric fields, which we do not consider here.

Here we have characterised both AM and FM noise on the RF chain and estimated the expected contribution to motional mode instability. Further characterisations of the motional mode stability using the ion as a sensor is discussed in section ??.

1.2.2 Trap DC Voltages

Static fields are required for reshaping the trapping psuedo-potential. This is useful for setting desired mode frequency splittings, for shuttling ions, and for compensating stray electric fields. As shown in figure 1.2, the *NPL* trap has 40 independently controllable DC electrodes. A further DC bias voltage can be applied directly to the silicon bulk. DC Voltage is supplied to the LCC through the following chain:



and embedded within the schematic 1.1. Two multi-channel DACs, known as *Fastino*, part of the *Artiq Sinara* hardware ecosystem, provide the initial 41 DC channels. The *Fastino* outputs are filtered by a passive cascaded RC low pass filters with resistances, $R = 2k$, and capacitances $C_1 = 2.2$ nF and $C_2 = 4.7$ nF⁸ yielding a -3 dB “cut-off” frequency of 7.4 kHz. The low pass filter circuit is shown in the above schematic, with the first stage on an external to vacuum PCB (Ext. Filt), and the final shunt capacitor on an in-vacuum PCB (Vac. Filt)⁹ within close proximity of the trap chip to mitigate RF pickup on the DC lines. The feedthrough to the vacuum system has 51 pins¹⁰ with non-DC pins used to provide ground connections. On the external filter board, one channel, corresponding to electrode *Ba co4*, has a capacitively coupled in AC signal. This originates from a DDS (*Urukul*) and is used to resonantly excite the motional modes of the ion chain for characterisation and calibration purposes (see section ??). The in-vacuum PCB is connected to the trap chip via a custom PEEK interposer with embedded *Fuzz buttons* vias¹¹.

The ion is confined in the “operation” zone, seen in figure 1.2 (c). The axial trapping is provided by electrodes *Ba dc5* and *Fr dc5*, where *Ba* and *Fr* correspond to the back and front plane of the trap respectively. The DAC can provide up to 10 V to these electrodes to produce an axial frequency of 1.6 MHz.

⁸For in-vacuum use. 1206 NPO 4.7nF 100V, SRT 1206A472JBCB

⁹Vacuum compatible PCB using Rogers RT6002. Filter boards were designed by Mariella Minder.

¹⁰*Allectra* 51-way Micro-D

¹¹*Custom Interconnects* Au/BeCu 0.025" diameter x 0.104" length Fuzz Button

1.3 Magnetic Field

Due to the fragility of the internal states of the ion (these are *literally* state-of-the-art magnetic field sensors), we must take great care in isolating the ion from a noisy environment. We do not have access to magnetic field insensitive transitions in $^{40}\text{Ca}^+$, therefore uncontrolled Zeeman shifts of the internal atomic energy levels must be suppressed through enforcing a stable external magnetic field. This magnetic field stability is required for long spin coherence times (see section 2.1.5), and for low error single- and two-qubit gates. A permanent magnet array of Samarium Cobalt¹², $\text{Sm}_2\text{Co}_{17}$, is used in a Helmholtz configuration to create a stable magnetic field of $\simeq 5 \text{ G}$ ¹³. Samarium Cobalt was chosen for its low temperature coefficient of remenance, $-0.03\%/K$.

The ion is shielded from unwanted external magnetic fields by a custom designed MuMetal enclosure¹⁴, with an internal dimension of $650 \times 650 \times 650 \text{ mm}$. The shielding consists of two layers of MuMetal shielding with internal shield thickness of 1.5mm and external shield thickness of 2.0mm. The factory quoted shielding factor at the center of the box is $\times 546$ for DC fields.

1.4 Vacuum System and Beam Geometry

Ultra High Vacuum (UHV) is required to prevent ion loss. However, UHV equipment is often bulky, and puts constraints on the access and visibility of the ion chain. Here, we describe the custom vacuum system, allowing for multiple beam access geometries including dual high numerical aperture (NA) objectives required for single addressed optical standing waves. The vacuum system was designed by Sebastian Saner, Chris Ballance, and Mariella Minder, and constructed by Sebastian Saner, Fabian Pokorny, and myself.

A residual pressure of $< 10^{-11} \text{ mbar}$ is desired. For this strict requirement, care

¹²*Bunting Magnets* E643SM and E644SM.

¹³Designed and constructed by Sophie Howarth.

¹⁴*Magnetic Shields* custom design.

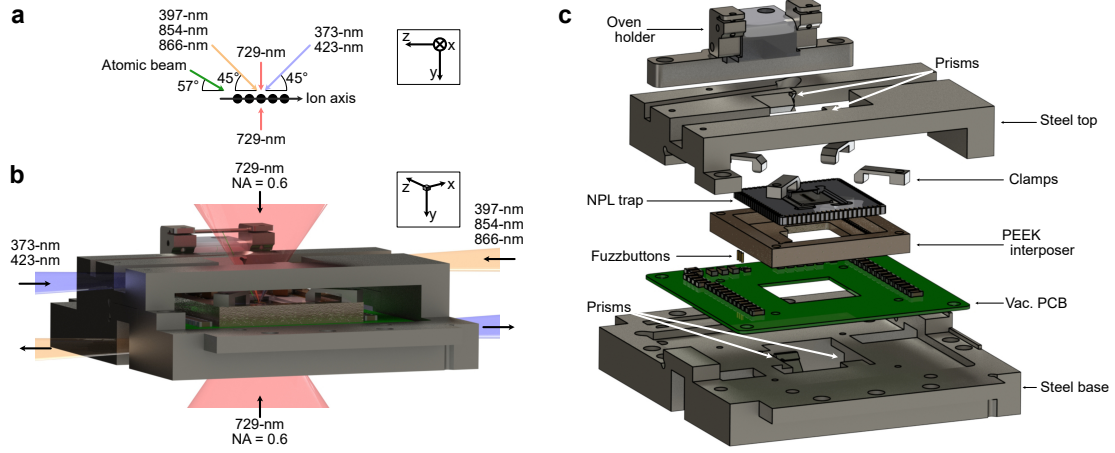


Figure 1.5: **a)** Laser and atomic beam geometries with respect to the ion chain. **b)** Compact trap housing with dual NA=0.6 access. The size of the high NA objectives restricts access for other beams, and so in-vacuum mirrors are used to redirect low-NA beams (373, 397, 423, 854, 866-nm) to the trap center. **c)** Exploded view of the trap housing with components labelled.

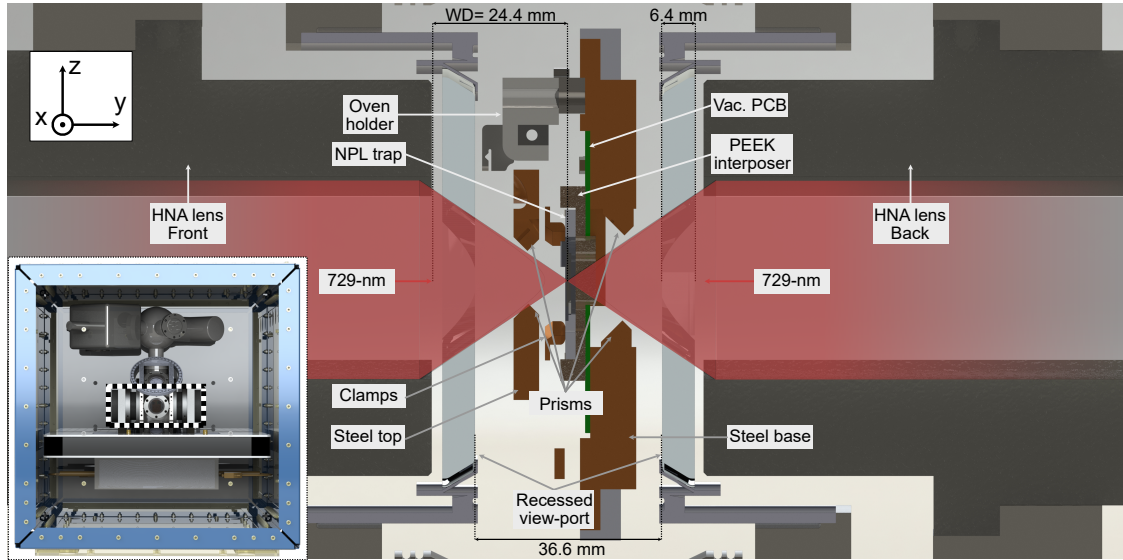


Figure 1.6: Cross section of the vacuum chamber and dual high NA objective sandwich. Lower right inset shows the vacuum chamber housed in the MuMetal enclosure, with dashed lines indicating the cross section region. Custom recessed CF-100 viewports are used to provide optical access for the high NA objectives (working distance = 24.4 mm).

must be taken in selecting in-vacuum materials, and thorough cleaning and baking procedures must be followed. A summary of tactics that were useful in the construction of the vacuum system can be found in [24, 25].

The system consists of a 6-inch spherical-octagonal experimental chamber¹⁵ connected to a spherical chamber with ion pump¹⁶, ion gauge¹⁷ and Titanium sublimator pump (TSP)¹⁸ attached. After the final system bake, a He leak test was performed and no external leaks were found. We do observe a gradual pressure increase that we suspect is either due to the in-vacuum optics, the optics epoxy adhesive¹⁹, or the PCB solder. Regardless, the ion pump and TSP maintain UHV at the desired $< 10^{-11}$ mbar (the TSP is fired every 4 weeks at 41 A for 60 seconds to provide a fresh Ti layer).

Figure 1.5 (a) shows the beam geometries with respect to the ion chain axis. The quadrupole 729-nm beam has high NA (NA = 0.6) access perpendicular to the ion chain axis, while 373-nm, 397-nm, 423-nm, 854-nm and 866-nm beams have low NA (NA < 0.05) access at an angle of 45° to the ion chain axis. The atomic beam from the oven is at an angle of $\sim 57^\circ$ to the ion chain axis. As such, there is a doppler shift of the atomic beam with respect to the 423-nm photoionisation laser, which is accounted for in the set laser frequency (given 12° angle between atomic and 423-nm beam, and 500 K mean temperature, expect +126 MHz frequency shift). Figure 1.5 (b) shows the custom trap housing including NA=0.6 access for the 729-nm beams, and in-vacuum mirrors (prisms) to redirect the low NA beams to the trap center. The low NA beams enter the housing via one of the four beam access channels parallel to the trap chip plane, and are reflected through the trap slit. The exiting beam is reflected by a second mirror out of the housing to prevent scattered light incident on the ion chain. Figure 1.5 (c) shows an exploded view of the trap housing with components labelled. The trap chip is interfaced to an in-vacuum PCB (described in section 1.2.2) via a custom PEEK interposer with embedded

¹⁵Kimball MCF600-SphOct-F2C8

¹⁶Agilent VacIon Plus 20 Pump

¹⁷Agilent UHV-24P Ion gauge

¹⁸Scanwel custom housing

¹⁹EPO-TEK 353ND

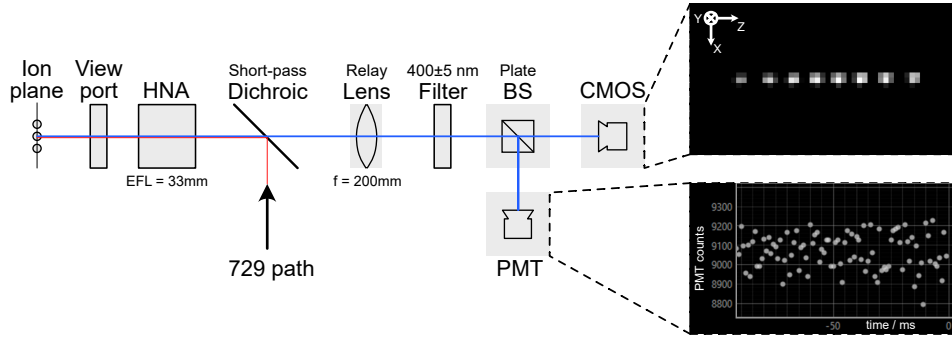


Figure 1.7: Optical path for fluorescent imaging of the ions using both a CMOS camera and a Photon-multiplier tube. Insets on right hand side show typical signal for: spatial resolution of eight ions in a linear chain (top), time resolution of detected photons from the ion chain (bottom).

Fuzzbutton vias (one for each DC and RF channel). The trap chip is held in place by 4 steel clamp arms, which attach to a steel base plate which provides mechanical stability and houses the lower in-vacuum prisms. A top steel plate houses the upper prisms. The tube oven is mounted on two steel clamps which are electrically isolated from the base plate by a PEEK spacer. The trap housing is fastened to the vacuum can by 4 steel arms attached to alternating side ports of the octagonal chamber²⁰. Figure 1.6 shows a cross section of the vacuum chamber and recessed dual high NA objectives. Two recessed CF100 viewports²¹ are required to bring the custom objectives²² to the designed working distance of 24.4 mm from the ion chain.

1.5 Imaging System

To image the ion chain, and readout the qubit states, fluorescent scatter of incident 397-nm light is collected. Here we briefly describe the imaging system, while section 2.1.2 describes the qubit readout procedure and fidelity. The imaging system is designed to provide high spatial resolution of the ion chain, while also providing a large field of view to accommodate long ion chains. The imaging system must also be compatible with the high NA objectives used for single addressed optical

²⁰Using *Kimball Groove grabbers* P/N 53-482610

²¹*UK Atomic Energy Authority* P/N VPR100015

²²*Photon Gear, inc.* Custom 18WD Atom Imager

standing waves. Figure 1.7 shows the optical path schematic of the imaging system, with both a high quantum-efficiency ($\simeq 70\%$ at 397 nm) CMOS camera²³ for spatially selective qubit readout on a multi-ion chain, and a photon multiplier tube (PMT)²⁴, for experiments requiring time resolved photon counting (e.g. micromotion compensation, section ??). Typical signals of both devices can be seen in right panels of figure 1.7. The optical path consists of a high numerical aperture objective²⁵ (effective focal length, $f = 33$ mm), $NA = 0.6$, a dichroic mirror²⁶ (transmission: 397 nm; reflection: 729 nm), a relay lens²⁷ ($f = 200$ mm), a 90 : 10 plate beamsplitter²⁸, and the camera and PMT.

The objective lens is custom designed to compensate for aberrations due to the recessed vacuum viewport, however, due to the large NA, accurate alignment of all degrees of freedom of the objective is critical. As such, the NA is attached to a 5-axis alignment stage²⁹, with z-axis focus controlled by a piezo micrometer³⁰ for fine adjustment of the ion focus with the MuMetal box closed (requires $< \mu\text{m}$ precision).

Given the chosen lens focal lengths, we expect a magnification of $\times 6$ from the ion plane to the imaging plane. With an expected ion separation of $5 \mu\text{m}$, and the camera pixel pitch of $6.5 \mu\text{m}$, we expect around 4.5 pixels between ions, which, if the system is well aligned, is more than sufficient for low cross-talk spatial readout of qubits.

1.6 Ca^+ Laser Systems

Singly-charged group 2 elements are popular in ion trap experiments due to their single outer electron resulting in Hydrogen-like energy levels. In this thesis we use $^{40}\text{Ca}^+$.

$^{40}\text{Ca}^+$ has no nuclear spin giving the (relatively) simple level structure shown in

²³Hamamatsu Orca-Fusion BT C15440-20UP

²⁴Hamamatsu Photon counting head H10682-210

²⁵Photon Gear, inc. Custom 18WD Atom Imager

²⁶**Thorlabs TODO**

²⁷Thorlabs Achromat Doublet, ACT508-200-A-ML

²⁸**Thorlabs TODO**

²⁹Newport ULTRAlign M-562-XYZ, M-36, SM-13

³⁰Newport CONEX-TRA12CC

figure ?? with the relevant laser transitions used in our apparatus are indicated. Details on the ion choice, and utility of the lasers is given in section ??, here we only describe the hardware and feedback control to provide this stabilised light. The relevant wavelengths are the following: The qubit is defined by the quadrupole transition at 729-nm. Access to other excited levels, outside of our defined two level system, are crucial for ion trap quantum logic to enable qubit readout via state selective fluorescence (397-nm, and 866-nm transitions), and state preparation via optical pumping (866-nm, and 854-nm transitions). Access to two additional transitions in neutral calcium for isotope selective ionisation, 423-nm and 373-nm, are also required [26].

This section begins with a description of laser frequency and power stabilisation techniques common to multiple laser systems used in this thesis. Following this, the specific systems are described, starting with the diode lasers, which drive all but the 729-nm transition. Subsequently, the Ti:Sapph 729-nm source is discussed. Finally, the global and single addressed beam arrangements for the 729-nm system are described.

1.6.1 Frequency Stabilisation

All lasers exhibit frequency noise due to unavoidable environmental coupling (e.g. temperature fluctuations, and acoustics), and to instabilities in control signals (e.g. voltage and current controls). For the lasers corresponding to dipole transitions used in this thesis, we are relatively lenient and require slow frequency excursions of < 1 MHz with respect to the qubit transition, and laser linewidths less than the natural linewidth of the dipole transitions, which are often multiple MHz. However, for the quadrupole transition at 729-nm, we require significantly more stringent frequency stability, with laser linewidths of < 100 Hz, and negligible phase modulation of the laser light over a few MHz of the qubit transition to ensure spectator transitions are not excited during qubit operations. Although the order of magnitude of the required frequency stability is different for the dipole and

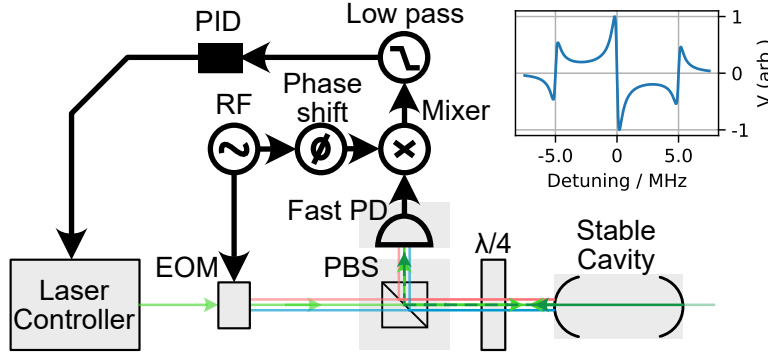


Figure 1.8: PDH optical and control schematic. Green optical path represents the carrier frequency, blue and orange lines represent the upper and lower sidebands respectively, generated by an EOM. The reflected light is directed onto a fast photodiode, and the signal is demodulated with the same oscillator signal used to drive the EOM, but delayed by some chosen phase. The signal is finally low pass filtered to recover the error signal, an example of which is seen in the upper right panel for a finesse $F \sim 4000$, a free spectral range of $FSR = 1.5$ GHz, and a modulation frequency of $f_{mod} = 5$ MHz.

quadrupole transitions, the concepts and techniques used to achieve this stability are similar. Generally, a closed loop consisting of measurement of the optical frequency of the laser and feedback to the laser control is required. In this work, stable external optical cavities are used for referencing the laser frequency. Specifically, we use the Pound-Drever-Hall (PDH) techniques to generate the error signal for the feedback loop. The PDH technique is well described in the literature [27–29], and so here we only give a brief overview of the technique to contextualise the required hardware introduced in the following sections.

The conceptual goal of the PDH technique is to measure the frequency difference between the laser and a resonant frequency of a stable optical cavity to generate an error signal, preferably linear over typical frequency excursions (while *locked*). This error signal can then be modified by some controller (typically PID), to directly infer how to tune the laser controls to minimise this error signal, often termed *locking* the laser to the cavity. Further, the error signal should be insensitive to amplitude noise on the laser light, to avoid amplitude to frequency noise conversion due to the feedback loop (in contrast, side-of-fringe stabilisation is sensitive to amplitude drifts).

Practically this scheme consists of phase modulation of the incident laser light, here by electro-optical modulators (EOMs), to generate sidebands, which are used as phase references to compare the phase of the incoming laser field to that of the phase of the field inside the cavity. Figure 1.8 shows a schematic of the PDH scheme, highlighting the beam paths and closed loop. The phase modulated light is directed onto the cavity, with the total return signal being some combination of the reflected sidebands, carrier, and the cavity leakage field, with the latter providing the phase of the electric field inside the cavity. The reflected light is directed onto a fast photodetector, and the signal is demodulated with the same oscillator signal used to drive the EOM, but delayed by some chosen phase. The resulting signal is low pass filtered to remove high frequency components, yielding the final error signal, an example of which is seen in the upper right panel of Figure 1.8. The central region is anti-symmetric and linear, centered on the desired lock point. Using a tuned controller for feedback to the laser controller, this provides a stable lock of the laser to the cavity.

As the cavity is used as a frequency reference, both a high finesse and stable optical lengths are desirable. The gradient of the PDH error signal is proportional to finesse, leading to increased signal to noise with higher finesse (see Ref. [28] for more details). Optical length deviations are minimised through mounting the cavity in a vacuum chamber, and taking care in isolating it from temperature fluctuations and vibrations.

Particular details of the PDH hardware and control loops are given in the following sections for each laser system.

1.6.2 Intensity Stabilisation

The light-matter interactions considered in this thesis are all proportional either to intensity or amplitude of the optical electric field at the ion's position. Intensity fluctuations over time therefore lead to fluctuations in the interaction strength, which are detrimental to the fidelity of quantum operations. Intensity noise originates

either from total beam power fluctuations, or due to beam pointing drift with respect to the ion position. The former is mitigated through a closed feedback loop between a pick-off photodiode signal before the ion monitoring the optical power, and the drive RF amplitude of an upstream AOM. For all beams, this optical pick off is one of the last elements before the beam enters the vacuum system to reduce the impact of lossy elements between the stabilise point and the ion position. The low-noise photodiodes used were designed by David Nadlinger (*QLNPD*), and the feedback system controller, known as *SUServo*, was written by Peter Drmota. Beam pointing drifts are mitigated through careful mechanical design of the optical paths for passive stability, and through automated intermittent beam pointing calibrations using the ion as a sensor (see section ??).

1.6.3 Diode Lasers

For transitions 397-nm, 854-nm, and 866-nm, we require light that is frequency stable to < 1 MHz level, well below the natural line widths of the dipole transitions, and power stable to < 1 μ W. All beam paths and relevant control loops for the diode laser systems are shown in figure 1.9. The laser diode heads³¹, controllers³², and splitterboards³³ (fig 1.9 (a)) are housed in a single 19" rack. The splitterboard distributes the light between two ion-trap experiments, the PDH lock cavity (fig 1.9 (c)), and a diagnostic wavemeter for checking wavelength and lock status³⁴. The splitterboard also houses an Electro-optic modulator (EOM) module for applying both offset and PDH sidebands for an offset PDH lock. The offset sidebands may be arbitrarily set (within the GHz range of the EOM) to provide a continuous choice of laser lock frequencies between cavity modes, rather than maintaining a fixed offset between the laser output frequency and the discrete cavity mode frequencies. For 397-nm and 423-nm, resonant free space EOMs³⁵ are used, for 854-nm and 866-nm

³¹*Toptica* diodes. 854/866: MDL DL pro; 397: MDL DL pro HP; 423: DL pro

³²*Toptica* DLC Pro with PDH boards

³³Designed by *OITG Cavities* group

³⁴*HighFinesse* WS7

³⁵*Qubig* PM7-VIS-25 and PM9-VIS-2.25

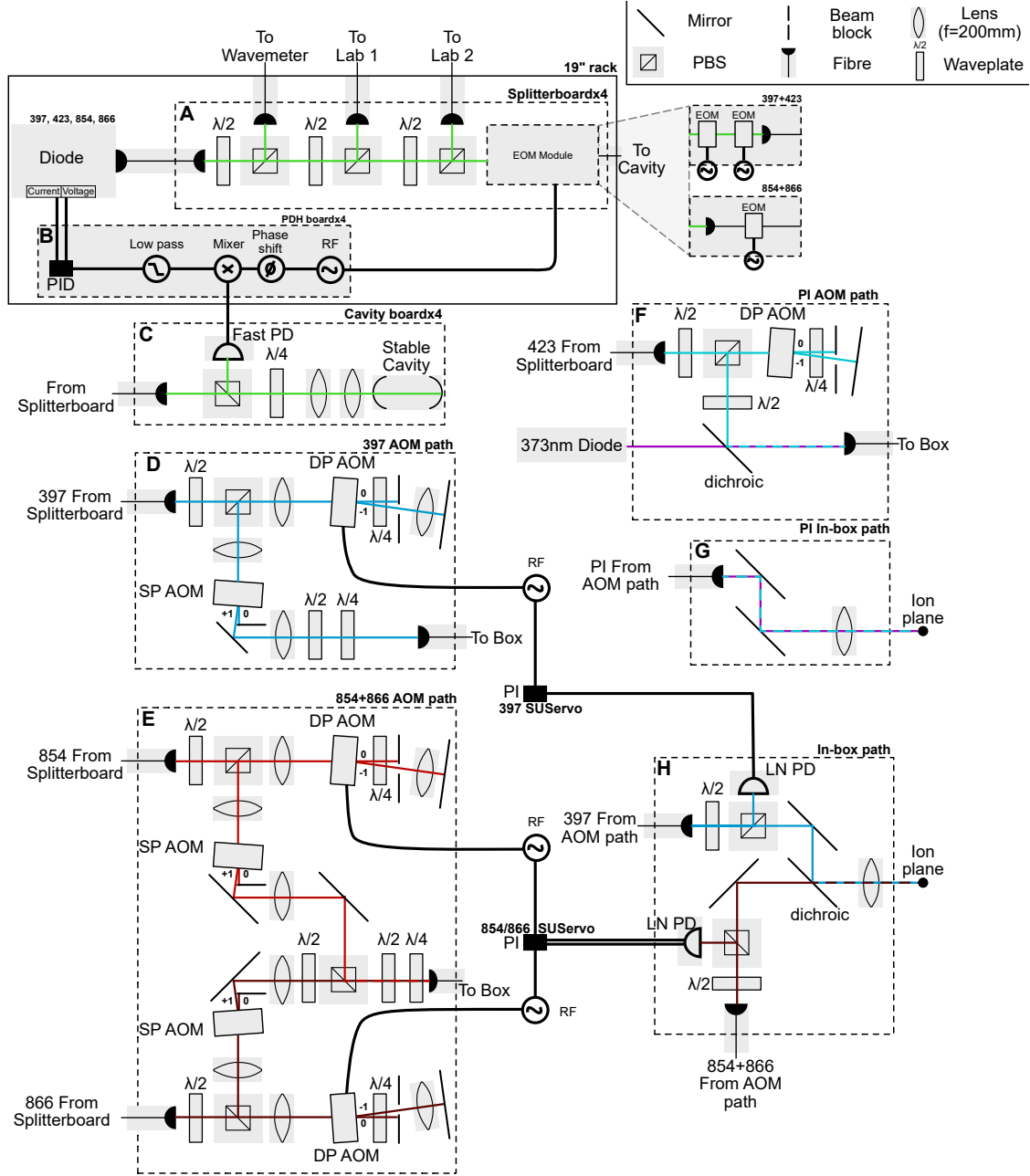


Figure 1.9: All beam paths for diode laser systems. Dotted boxes resemble modular optical paths with fibre and control lines in and out only. Frequency and optical power control loops are included.

broadband fibre EOMs³⁶ are used for convenience. The path on the reference cavity board (fig 1.9 (c)) consists of a four-bore reference cavity³⁷ under vacuum and temperature control, two lens for matching the collimated input beam to the cavity mode, and a beamsplitter and $\lambda/4$ waveplate for directing the reflected cavity light to a fast photodiode³⁸. The beatnote due to the carrier and PDH sidebands are mixed down and low-passed by the *Toptica* PDH module to produce the signature antisymmetric PDH trace. Gains for PID controllers of both the diode current and voltage are set to stabilise the optical frequency.

The light from the “laser” laboratory is carried to the “trap” laboratory using 16 m polarisation maintaining fibres³⁹. In the “trap” laboratory we use AOMs in a double pass configuration for both frequency shifting and power stabilising the light (fig 1.9 (d) and (e)).

As space inside the MuMetal enclosure is constrained, it is advantageous to couple multiple beams in a single fibre before the enclosure. The second stage photo-ionisation light, at 373-nm⁴⁰, is combined with the 423-nm beam by a dichroic (fig 1.9 (f)) before being coupled to a 16 m fibre that takes the light directly into the MuMetal enclosure. The respective refractive indices of fused silica for 373-nm and 423-nm differ to the extent that the two beams were separated by more than one waist at the ion plane after initial rough alignment. To ensure the two beams overlap at the ion plane, care was taken for the beam propagation to be perpendicular to both the lens and vacuum viewport window, and a parabolic mirror outcoupler⁴¹ (rather than a lens) was used inside the MuMetal box.

The 854-nm and 866-nm light is combined after their respective double and single pass AOMs, and coupled into a 3 m PM⁴² fibre carrying the light into the enclosure. Light from both the PI and the 397/854/866-nm paths are directed onto the ions via the in-vacuum prisms, discussed in section 1.4. The final mirrors prior to the

³⁶*EOSPACE* PM-0s5-05-PFA-PFA-850

³⁷*Stable Laser Systems* SLS-6010-1-4bore

³⁸*Thorlabs* PDA10A2

³⁹*Schafter-Kirchoff ...*

⁴⁰*Toptica* iBEAM-SMART-375-S.

⁴¹*Thorlabs* RC02APC-F01

⁴²*SK ...*

vacuum system are motorised⁴³, so that beam alignment to the ions can be optimised when the MuMetal enclosure is closed. Practically we find that these beams rarely need to be realigned (few months), unless there is a significant ambient temperature change of a few degrees.

1.6.4 Narrow Linewidth 729 Laser

Lasers are a key tool for creating the highly localised, strong electric field amplitudes and gradients needed to drive both carrier and sideband transitions of the trapped ion.

As shown in Figure ??, two sublevels within the $4S_{1/2}$ to $3D_{5/2}$ manifolds define the qubit. This is an electric quadrupole transition as $\Delta l = 2$. For the Calcium ion this transition is at 729-nm, and so a near resonance 729-nm laser is used to implement single- and multi-qubit gates (sections 2.1.3 and 2.3.2). This transition is also used after Doppler cooling for resolved sideband cooling to prepare the motional mode close to its ground state (as discussed in section 2.2.2). This transition is narrow linewidth due to the long lived $3D_{5/2}$ level, and so, for power efficiency, a narrow linewidth laser must be used. For the “fast entangling gates” use-case, high intensity electric-fields are required at the ion to drive sideband transitions, this will be discussed further in the single-addressing and two-qubit gate sections, but here it is sufficient to say we require >100 mW of light at the ion plane. Here we describe the 729-nm system consisting of a Ti:Sapph laser system pumped by an Nd:YAG 532-nm laser.

An *M2 Solstis* Ti:Sapph is pumped via 18 W of 532-nm light from a *Coherent Verdi-V* system to produce around 5 W of 729-nm light. The Ti:Sapph is engineered to operate with a stable < 50 kHz linewidth. Ti:Sapph crystals have broadband gain profiles which make them amenable in research environments to create frequency tunable laser systems. They also often have lower intrinsic phase noise than diode lasers at the same wavelength, which is favourable for our application. For single

⁴³*Newport Picomotor* piezo mirror mount P/N 8821

mode operation, the *Solstis* employs multiple intracavity frequency selective elements which consist of (in order of coarse frequency selectivity), a birefringent filter, a tunable Fabry-Pérot etalon, and the surrounding bow-tie cavity. To lock the tunable etalon to one longitudinal cavity mode, an actively stabilised “dither” phase-lock loop is used. This dither consists of periodically varying the etalon spacing with a frequency of around 20 kHz. This dither frequency directly phase modulates the light, which leads to the creation of sidebands which can interact with the ion causing errors in gates. This dither frequency (and harmonics of it), can be observed via composite pulse experiments on the ion, and are discussed in section 2.1.5.

Referencing the *Solstis* output with an external cavity and using PDH feedback can greatly suppress phase noise. Here an ultra high finesse cavity⁴⁴ is used, with a schematic of the 729-nm system shown in Figure 1.10. As opposed to the simpler controller implemented for the diode lasers, the 729-nm uses a cascaded PID lock with a fast feedback loop to the AOM situated after the *Solstis*, and an intermediate and slow feedback loop to piezos controlling the *Solstis* cavity length. With this system, a linewidth of $\ll 1$ kHz for the 729-nm light is expected. The electronics for this control loop are provided by *Stable Laser Systems* in the form of their *FPGA Servo* lock box. For an effective PDH lock, both short and long term stability of the reference cavity is required. To ensure the cavity is insensitive to the environment, it is made of an ultra low expansion material, and mechanically isolated from the environment by an acoustic dampening enclosure and a vibration isolation platform. The cavity is housed in a vacuum system to prevent refractive index fluctuations due to air, and to further reduce thermal coupling to the environment. The cavity is temperature stabilised at the zero crossing temperature of 30.6°C. Long term cavity drifts of 309(1) Hz/hr are measured over 125 days, seen in figure 1.11. These measurements use the ion at a known magnetic field as a frequency reference. Figure 1.10 displays the other beam paths for our 729-nm system. Some light is picked off and sent to a wavemeter to continuously monitor the frequency. However, the majority is coupled to two output fibres for our experiment and another within

⁴⁴*Stable Laser Systems* P/N 6020-4

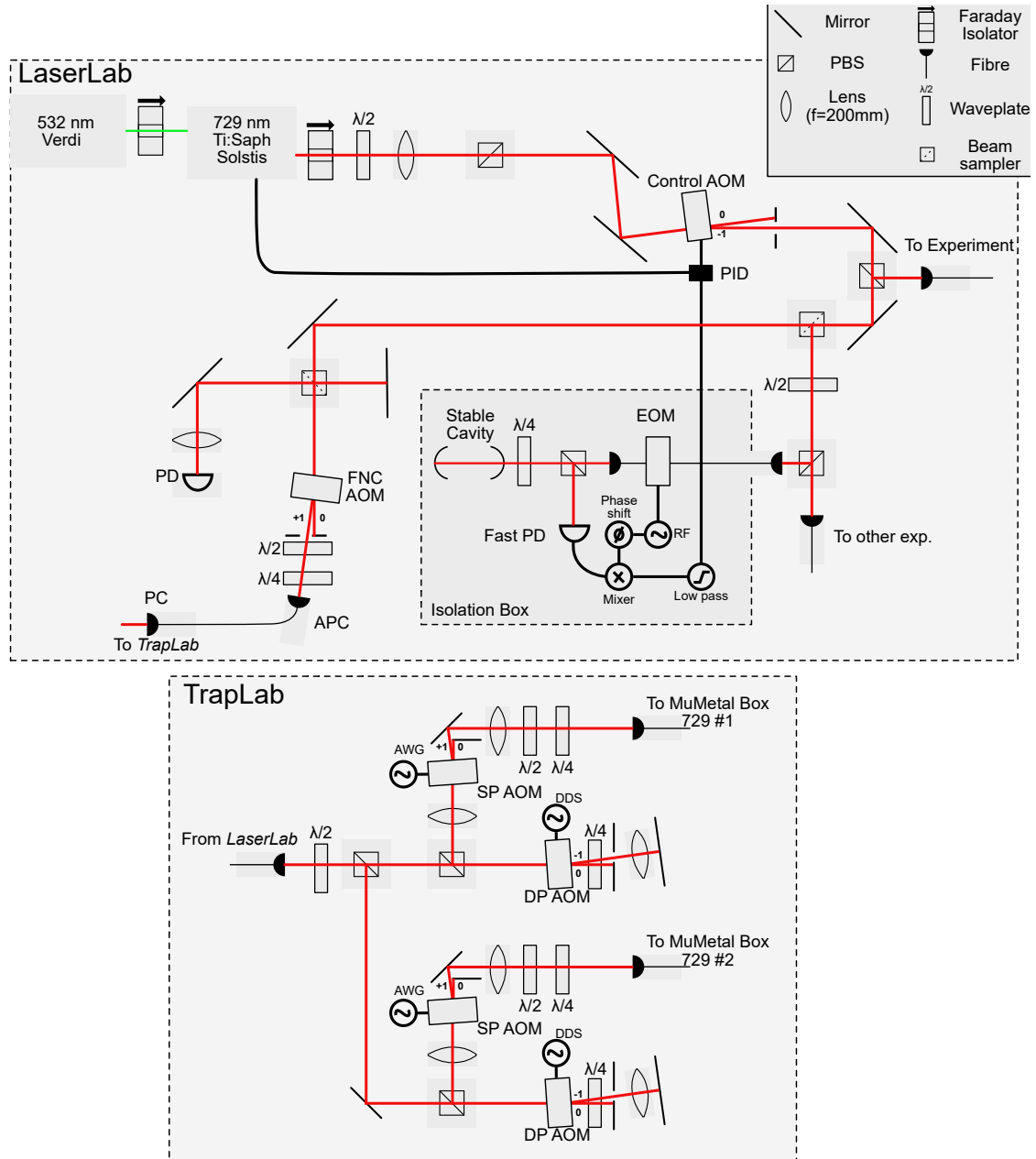


Figure 1.10: The 729-nm system. A Ti:Sapph laser tuned to 729-nm is pumped by a 532-nm source. Light is picked off at the first beam sampler to stabilise by PDH locking to a cavity. Light is distributed to two experiments, with the *FastGates* line including a fibre noise cancellation system.

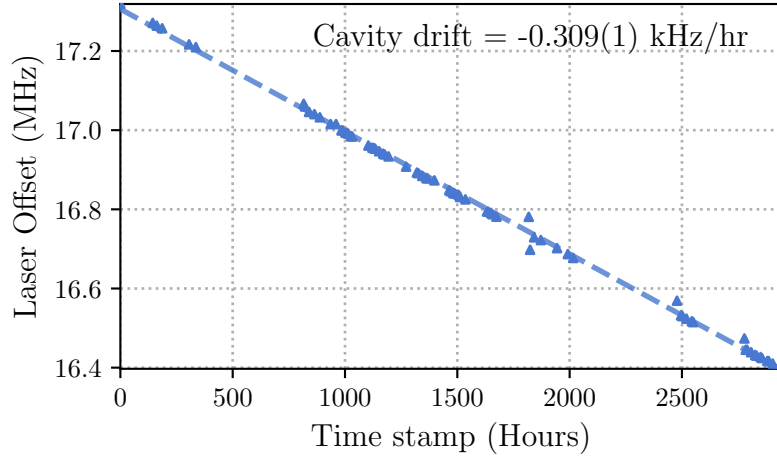


Figure 1.11: Cavity drift over 125 days measured by reference to the ion transition. The laser offset here is with respect to running all 729-nm AOMs at their central frequencies. Outliers are due to bad fits of the ion transition frequency, not due to changes in the cavity resonance.

the group. The 729-nm light is transported from a dedicated laser lab to a the trap apparatus lab by a 10 m single mode polarization maintaining fibre⁴⁵. The fibre introduces phase noise due to mechanical and thermal perturbations along the 10 m length. To remove this introduced noise, passive stabilisation in the form of thick foam tubing along the fibre length, as well as active stabilisation by a fibre-noise-cancellation (FNC) technique [30], is used⁴⁶. The efficacy of the FNC system is characterised in section 2.1.5 by measuring the spin coherence time of the ion.

1.6.5 Single-ion Addressing System

Selectively driving qubits and qumodes in a multi-ion register is a requirement for scalable QC. Our apparatus uses an optical single-ion addressing system to spatially select one or two ions in the chain. Spatial selectivity using narrow diameter beams is further advantageous as the focussed beam has greatly increased optical intensity

⁴⁵*SK ...*

⁴⁶FNC control consists of *Sinara Stabilizer* and *Pounder* mezzanine board with PID software developed by Ayush Agrawal.

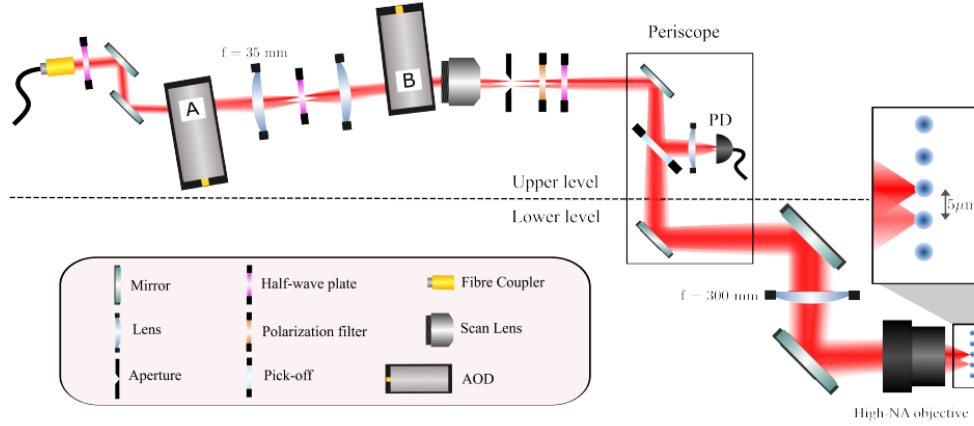


Figure 1.12: Schematic of the constructed compact single-ion addressing apparatus using crossed acousto-optic deflectors. The red beam represents the 729-nm field, with Gaussian waist changes (not to scale) indicating where in the optical train the beam is expanded or focussed. An approximately $1.5 \mu\text{m}$ waist is formed at the ion-position. This schematic is reproduced with permission from I. R. Overgaard transfer of status [31].

compared to a globally addressing field, and so drives much stronger interactions, and makes available much faster gates. A unique design feature of our system is the ability to produce an array of single ion addressing standing waves with individually contrrollable spatial phase, thus enabling any-to-any connectivity for fast engaging gates based on the scheme demonstrated in reference [3].

The design and construction of the single addressing system was led by Iver Øvergaard and, as of writing, has been in constant use in the experiment for around 1 year. Here we shall quote the design specification and relevant details of the single addressing system. In section 2.1.6 we present the characterisation data upon installation. Further technical information can be found in I. R. Overgaard’s Transfer report [31], available upon request.

Multiple methods for single ion addressing have been developed in the ion trap community, including waveguide arrays [32, 33], micro-electronic-mechanical mirror arrays [34, 35], spatial-light-modulators [36], and acousto-optic deflectors [37, 38]. Each method has its own advantages and disadvantages in terms of switching speed, resolution, ease of hardware integration, and scalability, and so the choice of

hardware is a matter of experimental preference, and constraints from the remaining ion-trap apparatus. We require:

- Addressing of up to two ions in a ≈ 10 ion chain with low crosstalk to neighbouring ions.
- Fast reconfigureability, $\approx 1 \mu\text{s}$, of the beam position within a single experimental shot (i.e. between different gates within a single circuit execution).
- Good optical power efficiency to ensure high intensity electric fields at the ion.
- Flexibility in beam positioning to allow for fine tuning of the overlap of two counterpropagating beams used to create the standing wave.
- Individual phase control of each addressed beam to stabilise the standing wave phase at the ion plane.
- Beam position stability to avoid power fluctuations.
- A compact design to fit within the magnetic shielded region of the experiment.
- Good telecentricity (fixed beam k -vector across the scanned area) to avoid unwanted travelling wave component in a standing wave gate scheme.
- Zero frequency shift of the spatially tunable beam at the ion plane.

Crossed Acousto-Optic Deflectors (cAODs) are a popular choice for single ion addressing, offering fast rise times ($< 1 \mu\text{s}$), and good resolution over a large scanning area (≈ 500 resolvable spots with 2° angular range) with zero frequency shift along the addressing axis. AODs operate through the same mechanism as AOMs, but are optimised for beam deflection rather than frequency modulation. As an addressing system cAODs have poor scaling of power efficiency with the number of addressed spots, and so are not ideal for large scale systems. As we are interested in addressing ≤ 2 ions at a time, this is not a significant issue for our system.

Figure 1.12 shows a schematic of the constructed addressing system. Due to

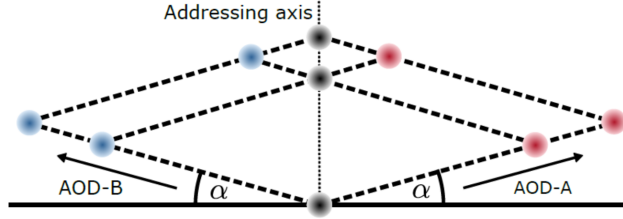


Figure 1.13: Spots generated by cAOD with wedge angle α , when supplying two frequencies to each AOD. The black points have no over-all frequency shift, and are aligned to the ion-chain. This schematic is reproduced with permission from I. R. Overgaard transfer of status [31].

space constraints within the magnetic shielded region, the optical elements are located on two height levels with a periscope steering the beam between them. The beam is delivered via single-mode polarisation maintaining fibre to the upper level where it is steered through the first AOD, with beam angle optimised for maximum diffraction efficiency into the -1^{st} order. Two lenses⁴⁷ in a $4f$ configuration re-image the deflection origin of the first AOD⁴⁸ onto the second AOD with the second AOD angled for maximum diffraction efficiency into the $+1^{st}$ order. The cAOD arrangement is termed “crossed” as the second AOD is rotated by some angle α with respect to the first, so that the deflection from the second AOD has some orthogonal component to that of the first. When the two AODs are driven with RF tones of the same frequency, f , the high efficiency diffracted order combinations are: $(0,0)$, $(-f,0)$, $(0,+f)$, $(-f,+f)$. The last combination of $(-f,+f)$ has zero overall frequency shift yet retains some frequency dependent deflection angle, which is suitable for addressing ions in a 1D line. When two frequencies, f_1 and f_2 , are supplied to the cAODs, two spots are formed at the ion plane. This arrangement is required for addressed two-qubit entangling gates. The spot configuration when two frequencies are present is shown schematically in figure 1.13. In this case, there are two spots close to the ion chain with frequencies of $f_x = \pm(f_1 - f_2)$ compared to the zero-frequency shifted spots aligned to the ion-plane. These nearby spots can lead to unintended crosstalk errors, particularly when addressing neighbouring ions.

⁴⁷ **P/N**

⁴⁸ **P/N**

The choice of wedge angle α then acts as a trade-off between total scan range of the steerable beam, and the crosstalk due to the off-ion-axis beams, with a decreasing α effectively further displacing the off-axis beams from the ion chain, as shown schematically in figure 1.13.

To translate this deflection angle into some physical displacement at the ion plane, a telecentric scanning lens⁴⁹ is used. The telecentricity ensures that the beam k -vector is fixed across the scanned area, which is important for standing wave gate schemes to avoid unwanted travelling wave components. At the intermediate focus after the scanning lens, the unwanted orders are blocked by an adjustable slit aperture and the beam polarisation is cleaned and selected using a polarisation filter and half-wave plate. A plate beam splitter⁵⁰ is used within the persicope to pick off $\approx 5\%$ of the light for monitoring on a photodiode⁵¹. This photodiode is used for an intensity-stabilising feedback loop as discussed in section 1.6.2. The scan lens, in combination with a relay lens, forms a beam expander to increase the beam NA for the final objective lens to focus to a beam waist radius of $\approx 1.5 \mu\text{m}$ at the ion plane.

Characterisation of the system (details given in section 2.1.6) determine that experimentally the addressing beam is focused to a Gaussian waist of $1.50(1) \mu\text{m}$, measured through Rabi-oscillations while scanning the beam position across a single trapped ion. The same beam was then tested in a five-ion chain, where the average nearest-neighbour intensity crosstalk was measured to be $4.16(2) \times 10^{-4}$, with crosstalk defined as the relative intensity at the probe position compared to the addressed position.

1.6.6 Global Addressing System

The single addressing provides the utility of spatially selecting a few ions to drive with the 729-nm light, however, the power efficiency scales as $O(n^2)$, with n being the number of addressed spots. As such, we limit the single addressing to simultaneously

⁴⁹**P/N**

⁵⁰**P/N**

⁵¹**P/N**

illuminate only one or two ions at a time. It is therefore convenient to also have a global beam that has dimensions larger than the ion chain to address all the ions in parallel. This global branch is currently counterpropagating to the single addressing path, and shares the same optical path as the imaging system (see figure 1.7). To increase the beam intensity at the ions, the global branch uses an anamorphic prism pair⁵² to elliptically shape the beam, with a 4:1 aspect ratio, with major axis along the ion chain direction. The beam is focussed onto the ion plane using an $f = xx$ lens⁵³, resulting in a major axis beam waist radius of $52(3) \mu\text{m}$ ⁵⁴. As with the other beams directed to the ion, we use a final piezo controlled mirror⁵⁵ for fine alignment and control when the MuMetal enclosure is closed.

1.7 Control System

Could tie up loose end here of control FPGA blocks.

1.8 Ion Loading

Atoms are provided to the trapping region by heating Calcium flakes housed in a resistive tube oven with the oven mount shown in figure 1.5. A thermocouple is tag welded onto the tube oven to provide a temperature reading for current feedback for fast stabilisation to the target temperature of $\simeq 200^\circ\text{C}$. A steady state of $\simeq 2 \text{ A}$ across the 0.5Ω load of the oven tube is required to maintain 200°C . The tube oven has a pinhole that is aligned to the ion flux aperture, visible in figure 1.5 with atom flux shown in green, and the atomic shield pinhole shown in figure 1.2. These apertures were included in the design to improve the operational lifetime of the trap by ensuring neutral Calcium metal cannot accumulate on the trap surface and cause shorts between RF, ground, and DC electrodes. The aperture design *luckily* failed. It failed as we observe atomic flux at all electrode locations suggesting the

⁵²ThorLabs PS883-B

⁵³ $\mathbf{P/N}$

⁵⁴The beam profile was measured by extracting per-ion Rabi frequencies while driving on resonance quadrupole transitions (see section 2.1.3) and assuming a Gaussian cross section.

⁵⁵ThorLabs POLARIS-K2S2P

aperture is too large. It was *lucky* as the planned trapping region of *dc10* became non-viable after electrode *Fr dc11* developed a continuity fault. As such, ions are loaded at *dc5*.

The neutral atom flux is ionised using a two-step photoionisation scheme [26] that can selectively load the desired isotope of Calcium. Natural abundance Calcium, with typical isotope proportions of 96.9% : ^{40}Ca , 2.09% : ^{44}Ca , 0.647% : ^{43}Ca [26], was appropriate as we aim to exclusively load $^{40}\text{Ca}^+$. The first stage involves a frequency stabilised 423-nm beam that couples the $4s^2 \rightarrow 4s4p$ transition in neutral calcium. The natural linewidth of this transition ($\simeq 35$ MHz) is much smaller than the isotope shifts between abundant isotopes of Calcium (typically > 100 MHz shifts), as such, through frequency selectivity, only the desired isotope can be excited. The second ionisation stage couples $4s4p \rightarrow \text{continuum}$ through a 373-nm UV laser source (although the exact wavelength does not matter as long as < 389.8 -nm). We practically see no loading of other Calcium isotopes, and only see rare non-fluorescent ions (once every few weeks) after collisions with residual gas (we suspect this is CaH^+).

The 373-nm laser causes charge build-up on the trap electrodes, which can affect the trapping potential, and so a physical shutter, controlled by a TTL signal, is used to block the beam when not actively loading ions.

1.9 Outlook

Does this section need a bit of a closing word?

1.9.1 Limitations

2

Calibration and Characterisation

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Before we can proceed with our planned experiments involving the motion and spin of the ions, the apparatus must be characterised. This characterisation serves two purposes: it allows us to benchmark the performance of our system, and it reveals any current limitations that must be addressed before we can successfully carry out the experiments.

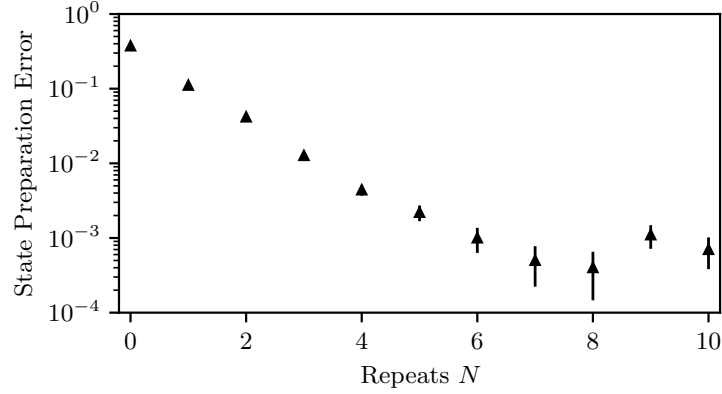


Figure 2.1: State preparation error as a function of the number of state preparation pulses, N . The error is defined as population not in the $|\downarrow\rangle$ state after the preparation sequence has been applied. The error is reduced to below 10^{-3} after $N = 7$ pulses, which is used in our experiments.

We split this chapter into interactions involving only the spin, then those focussed only on the motion, and finally we describe spin-motion interactions. This chapter demonstrates the full control required for discrete-variable quantum computation. Section ?? in chapter ?? applies the spin-dependent motional interactions demonstrated here for continuous-variable readout, and to realise the remaining non-linear and two-mode operations required for full hybrid CV-DV control.

2.1 Spin

Discrete variable quantum computing consists of control over many two-level systems, which we refer to either as spins, or qubits. In this section the experimental methods used to prepare, manipulate, and measure these spins are described, and the performance of these operations are benchmarked.

Parameter	Value
729-nm laser power	1000 μW
729-nm pulse duration	5 μs
854-nm laser power	60 μW , $250 \times I_{\text{SAT}}$
854-nm pulse duration	40 μs
866-nm laser power	15 μW , $\sim 70 \times I_{\text{SAT}}$
866-nm pulse duration	30 μs

Table 2.1: Experimental parameters used for state preparation of the ion into the $|\downarrow\rangle = |4S_{1/2}, m_j = -1/2\rangle$ state. All lasers frequencies are set to be resonant with the corresponding transition frequencies.

2.1.1 State Preparation

To utilise two levels of the ion as a qubit, the electronic state must first be prepared into the qubit manifold. Here, the ion is prepared into the $|\downarrow\rangle = |4S_{1/2}, m_j = -1/2\rangle$ state. This is done via optical pumping by pulsing the quadrupole transition $|4S_{1/2}, m_j = +1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$, followed by deshelling pulses using the 854-nm, 866-nm lasers on resonance. The 729-nm laser pulse duration and intensity is set to invert the populations, i.e. perform a π -pulse. Details for laser parameters are summarised in table 2.1. These three pulses are repeated N times sequentially. Figure 2.1 shows the state preparation error with increasing N . The error is found by applying a π -pulse on the state preparation transition directly after the state preparation sequence and measuring the population in the $|\downarrow\rangle$ state, $P_\downarrow$. Any population not in $P_\downarrow$ indicates either residual population in the $|4S_{1/2}, m_j = +1/2\rangle$ state, or trapped in the metastable D levels. After $N = 7$ pulses there is a state preparation error of $< 10^{-3}$. The state preparation error is likely limited by accidental excitation of the $|4S_{1/2}, m_j = -1/2\rangle$ state to the $D_{5/2}$ level. This may be due to off-resonant excitation of another transition, or due to spectral purity of the 729-nm laser.

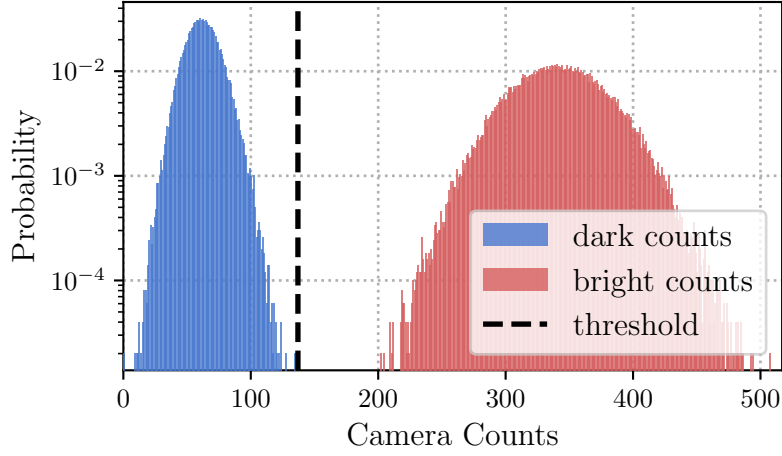


Figure 2.2: Readout histogram with 100,000 individual measurements for bright (red) and dark (blue) populations. The camera counts are presented as a histogram and fitted to Poissonian distributions to calibrate a threshold for discerning if the ion is bright or dark with the lowest probability for false measurement. Dark counts are mainly due to 397-nm light scattered from the nearest trap electrodes. With 100 μ s readout duration, and laser parameters set to Doppler cooling parameters, a threshold of 137 photons is found to be sufficient to discern the two populations.

Parameter	Value
397-nm laser power	3 μ W, $1 \times I_{\text{SAT}}$
397-nm laser detuning	-17.5 MHz
866-nm laser power	15 μ W, $\sim 70 \times I_{\text{SAT}}$
866-nm laser detuning	0 MHz
Doppler cooling duration	1 ms
Readout duration	100 μ s
Camera threshold	137 counts

Table 2.2: Experimental parameters used for Doppler cooling and state-selective fluorescence of the ion.

2.1.2 Measurement

To measure the qubit state, the 397-nm and 866-nm lasers are applied, and the scattered 397-nm photons are imaged by the objective onto a camera (see section 1.5). 397-nm light will only be scattered if the qubit is in $|\downarrow\rangle$. A typical image generated by the camera is seen in figure 1.7. A small window of pixels is set around each ion (typically a 3×3 px region), and the photon counts in this region are summed. Use

of a high numerical aperture ($\text{NA} = 0.6$) objective greatly increases the collection efficiency. This efficiency allows the 397-nm beam to be red-detuned to Doppler cooling parameters (see section 2.2.2), while still maintaining low readout duration (100 μs) and low readout error. Cooling during readout increases the experiment's repetition rate by reducing the amount of dedicated cooling time required.

The parameters used for readout are summarised in table 2.2, and a typical histogram of readout counts for one ion can be seen in figure 2.2. The readout threshold is calibrated by taking bright counts with the 397-nm and 866-nm lasers on, and dark counts with the 397-nm laser on and the 866-nm laser off. This assumes dark counts are due to 397-nm scattering from the trap electrodes. Fitting Poissonian distributions to the bright and dark measurements, an optimal count threshold is determined, shown by black dashed line. Assuming Poissonian statistics this threshold would give an expected statistical readout error of $\epsilon_{\downarrow} = \epsilon_{\uparrow} = 6 \times 10^{-9}$, where $\epsilon_{\downarrow, \uparrow}$ are the expected error associated with measuring bright or dark. This measurement ignores errors due to the finite lifetime of the metastable $3D_{5/2}$ level, which dominate $\epsilon_{\uparrow}$. The expected error due to the finite lifetime, τ , is $\epsilon_{\text{decay}} = 1 - e^{-t/\tau}$, where t is the readout duration. For $^{40}\text{Ca}^+$, $\tau \sim 1.1$ s, $\epsilon_{\uparrow} \approx \epsilon_{\text{decay}} = 10^{-4}$. Assuming the metastable lifetime is limited only by spontaneous decay overlooks leakage from the 854-nm deshelving beam, which could significantly shorten τ [32]. Further characterisation of the combined state-preparation and measurement fidelity is provided in § 2.1.4.

2.1.3 Single-Qubit Rotations

Single-qubit rotations driven by optical fields, as described in § ??, along with two-qubit gates are required for universal discrete variable quantum computation.

We divide errors affecting single-qubit gates into two categories: unitary, and non-unitary. The unitary error component is labelled “systematic” if there exists simple (and accessible) control parameters to suppress such errors. We outline in the next section robust phase estimation (RPE), the calibration routine used to

reduce systematic errors. In contrast, reducing non-unitary errors require improved isolation between the qubit and environment, or suppressed noise on the single-qubit-driving field. At the end of this section we characterise the performance of the single-qubit gates using randomised benchmarking.

Robust Phase Estimation

Many gate calibrations can be formulated as phase estimation tasks. A single-qubit rotation may be generally represented by $R_{\vec{n}}(\theta)$, where $\vec{n}$ is some given axis around which the state evolves with rotation angle $\theta = \omega t$, where the angle is accumulated over time by rate ω . For a gate we specify this axis and total angle, with deviations to the ideal setting being a systematic error.

Suppose we can set axis $\vec{n}$, and the interaction duration t arbitrarily well, the task of calibration is then limited to how precisely we can measure ω . We further assume some statistical uncertainty for a practical phase measurement, σ_θ , that is independent of the magnitude of θ . The uncertainty of angular frequency $\sigma_\omega = \frac{\sigma_\theta}{t}$ is thus inversely proportional to duration t — longer probe durations provide more information. This improved signal to noise is understood by the coherent accumulation of phase over time due to an underlying ω .

However, there are two issues with going to large t : first, incoherent errors also accumulate, attenuating the measured signal, and violating our assumption of a fixed statistical uncertainty σ_ω . Second, quantifying θ for a fixed duration t is ambiguous — the measurement only provides the wrapped angle, θ_w , for example in the interval $(-\pi, \pi]$. The first limitation restricts what practical maximum t we can measure, heuristically often set to the $\frac{1}{e}$ -coherence time, τ . The second limitation may be mitigated through supplemental measurements. The RPE protocol provides an algorithm for determining which branch the measured θ is on through an optimal number of supplemental measurements [39].

The protocol is as follows:

1. Select some duration t_0 such that $\theta_0 = \omega t_0 \in (-\pi, \pi]$, and determine parameter $k_{\max} = \left\lfloor \log_2 \left(\frac{\tau}{t_0} \right) \right\rfloor$, to set the maximum probe duration, $t_{\max} = 2^k_{\max} t_0$, below the coherence time, τ .
2. For $k \in \{0, 1, \dots, k_{\max}\}$, measure total phase $\theta_{k,\text{tot}}$ wrapped to $(-\pi, \pi]$.
3. Infer candidate base-duration phases for each k as $\theta_{k,0}^j = \frac{\theta_{k,\text{tot}} + 2\pi j}{2^k}$ for $j \in \{0, 1, \dots, 2^k - 1\}$.
4. The branch for $\theta_{k,0}$ is then recursively selected to be the closest $\theta_{k,0}^{(j)}$ to the determined $\theta_{k-1,0}$.
5. Angular frequency, ω , is inferred from $\omega = \frac{\theta_{\max,0}}{t_0}$ with uncertainty $\sigma_\omega = \frac{\sigma_0}{2^k t_0}$ where σ_0 is the phase uncertainty from a measurement at t_0 .

We use this protocol for both frequency (ω) and gate-duration, t , calibrations, which we outline below.

Transition Frequency Calibration

Rotations around the $\hat{\sigma}_z$ axis, correspond to a relative detuning, δ , between the qubit precession and a the reference frequency of a near resonant drive field, given by the idle Hamiltonian, $\hat{H}_z = -\hbar \frac{\delta}{2} \hat{\sigma}_z$. The phase accumulated due to the application of $\hat{H}_z$ for a fixed time, T , is $\theta = \delta T$. This phase may be measured through RPE using Ramsey spectroscopy. The sequence starts with an initial $R_x(\frac{\pi}{2})$ pulse, then a delay with duration $2^k t_0$, followed by a final $R_\phi(\frac{\pi}{2})$ pulse with variable rotation axis $\vec{n} = \cos(\phi)\hat{\sigma}_x + \sin(\phi)\hat{\sigma}_y$. Varying the final axis angle, ϕ , for a given probe duration reveals the familiar Ramsey fringes, with phase offset given by θ . For sample-efficient data collection, the phases $\phi \in [0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}]$ are chosen, which provide enough information to determine the phase angle in the xy -plane, amplitude, and any asymmetric qubit readout offset of the measured fringes.

By this method, we calibrate the reference laser-frequency with respect to the bare-qubit frequency. The Stark shifts due to the two $\frac{\pi}{2}$ -pulses will give some systematic offset between the bare-qubit frequency and the calibrated laser frequency, however

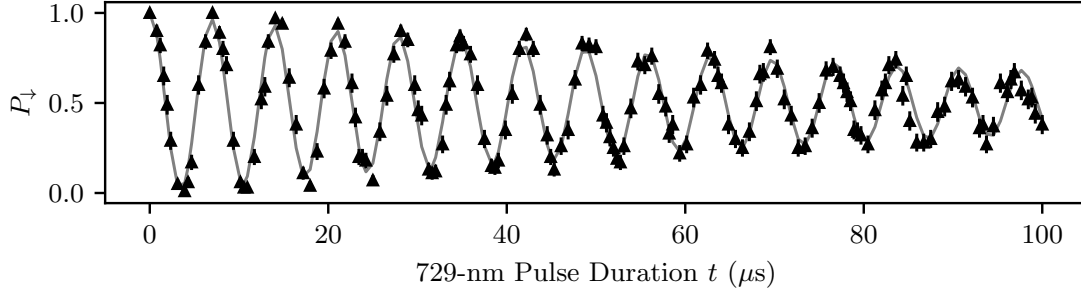


Figure 2.3: Rabi oscillations due to on-resonant 729-nm beam. Estimated probability $P_{\downarrow}$ at each time step is average of 200 individual experimental shots. The dotted line is a fit using decaying oscillation model given by equation 2.1.

are negligible for long Ramsey probe durations.

Resonant $R_x(\theta)$ gates are then driven by setting the relative detuning to $\delta = 0$. This works well for gates using low optical intensity, however, if high intensity is required (e.g. for motional-sideband pulses), a further calibration of the beam-induced Stark-shift is required. In the case that the shift is dominated by far-off-resonant dipole transitions, the described Ramsey experiment may be simply modified to include the optical dressing field to be present during the delay period. The dressing field is detuned sufficiently to not drive resonant interactions, but still provide near-identical Stark-shifts on the qubit transition. Using a Hahn-echo sequence with the dressing field present only in one of the arms allows for extracting the dressing field induced Stark shift while being insensitive to any systematic frequency offsets between the bare-qubit and the calibrated resonant $\frac{\pi}{2}$ -pulses.

Gate Time Calibration

With the resonant frequency well calibrated, the pulse durations for various gates must be set. Here we outline the calibration routine for a $R_x(\frac{\pi}{2})$ -gate, although this may be adapted for gate angles correspond to integer fractions of 2π .

We use a modified version of the RPE protocol to determine optimal gate durations. This protocol converges with a sufficiently close initial value for pulse duration $T_{\pi/2}$, and so a simple calibration using resonant Rabi oscillations is first described.

The qubit is prepared in the computational eigenstate $|\downarrow\rangle$, followed by an on-resonant 729-nm carrier pulse inducing $R_x(\Omega t)$ rotations. The final state population is measured as a function of pulse duration to observe familiar Rabi oscillations. Figure 2.3 displays these oscillations when using a global-addressing beam with a total power of 1 mW. Here each point corresponds to the average of 200 samples. The Rabi frequency, Ω , and decay rate, λ , is determined through fitting the recovered populations to the model,

$$P_{\downarrow} = \frac{1 + e^{-\lambda t} \cos(\Omega t)}{2}. \quad (2.1)$$

In this case, the Rabi frequency is found to be $\Omega = 2\pi \times 0.1433(1)$ MHz, and the decay rate $\lambda = 0.0104(6) \mu\text{s}^{-1}$. From this a first-guess pulse duration of $T_{\pi/2} \approx 1.75 \mu\text{s}$ can be estimated. This is a sufficient initial value to calibrate further, however it does not account for any amplitude ramps or dead-time of the optical-pulse compared to the digital control signal.

RPE provides a higher-precision calibration, that may account for non-square-pulse profiles. The protocol is simply adapted from the RPE method described above by setting the initial base probe as a single $R_x(\frac{\pi}{2})$ -pulse, with duration $t_0 = T_{\pi/2}$. The pulse is repeated 2^k times up to k_{max} (inferred from characteristic decay time $\tau = \frac{1}{\lambda}$) applications of the $R_x(\frac{\pi}{2})$ -pulse. The phase angle in the yz -plane is determined through applying an additional $R_x(\theta_f)$ rotation with angle $\theta_f \in [0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}]$, before measurement. The final base phase accumulated, θ_0 , is determined after applying the phase unwrapping protocol given above. Any deviation between this measured phase and the target, $\frac{\pi}{2}$, may be subsequently calibrated by rescaling the original pulse in time by factor $s = \frac{\pi}{2\theta_0}$.

This method, along with the transition frequency calibrations given above, provide an efficient method to calibrate systematic offsets of single qubit rotation gates driven through on-resonant carrier excitation. Next we discuss randomised benchmarking, a characterisation technique that provides a measure of remaining incoherent and systematic errors for the calibrated gates.

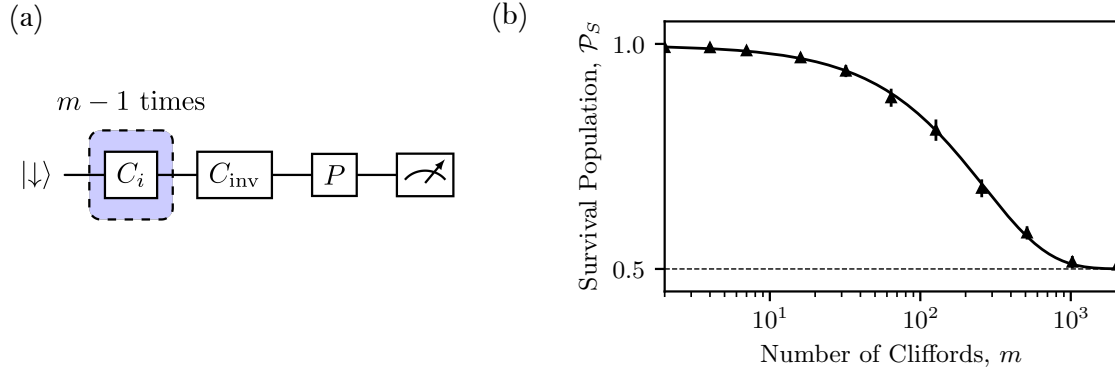


Figure 2.4: Standard randomised benchmarking applied to a single qubit. Panel a) shows the required circuit for a random sequence of Clifford gates, C_i , of total length $m-1$, with a final inverting Clifford, C_{inv} , and a random Pauli gate, P , before measurement in the computational basis. Panel b) shows the survival probability, $\mathcal{P}_S$, versus Clifford sequence length, m . Each point is found by averaging over $r = 50$ and $s = 50$ repeats for each m . The solid line is a fit using the decay model given by equation 2.4, the dashed line is the expected asymptote for a fully depolarised single qubit.

2.1.4 Single-Qubit Randomised Benchmarking

A further characterisation protocol is required to quantify the performance of single-qubit gates under both systematic and incoherent noise. A scalable class of methods, prevalent in quantum computation, is that of randomised benchmarking (RB). We will give a short description of RB here, but further details of the goals of characterisation, and complementary techniques to RB are discussed in chapter ?? § ??.

The purpose of RB is to simplify a general completely positive, trace preserving (CPTP) error channel by enforcing the symmetry of a chosen group on the channel [40]. The general error channel is described by the map acting on the space of density matrices,

$$\mathcal{E}(\rho) : \mathbb{C}^{d \times d} \rightarrow \mathbb{C}^{d \times d} \quad (2.2)$$

where d is the Hilbert space dimension. The symmetry is enforced on the channel through twirling — averaging a channel $\mathcal{E}$ over a group $\mathcal{G}$. A treatment of the action of twirling on a CPTP channel is given in § ?? in the context of k -copy RB,

an extension to standard RB methods for a single-qubit [41].

The resulting simplified channel is then quantified to obtain metrics of the performance of the executed circuit. This protocol thus can reduce the problem to measuring $O(1)$ parameters, rather than the $O(d^4)$ parameters required for a general quantum channel acting on a Hilbert space with dimension d . Furthermore, RB is particularly robust to state-preparation and measurement errors (SPAM), allowing for the isolation of errors originating from imperfect SPAM, to errors originating from the qubits idling or from the execution of gates.

Practically RB protocols are simple to implement, approximating the twirling procedure to the generation and execution of an ensemble of random gate sequences of varying lengths, m . We focus on the “standard” RB sequence, enforcing the single-qubit Clifford group symmetry onto a single-qubit error channel [41]. The aim of standard RB is to twirl over the single-qubit unitary group, $SU(2)$, which takes the Haar measure, and transforms single-qubit error channels to highly-symmetric depolarising type channels, fully expressed by a single parameter, λ ,

$$\mathcal{E}_{dp}(\rho) = (1 - 2\lambda)\rho + \lambda I_2 \text{Tr}(\rho), \quad (2.3)$$

where $\mathcal{E}_{dp}$ is the resulting twirled channel, and I_2 is the 2×2 identity matrix. Unfortunately, $SU(2)$ is continuous, as such it is impractical to average over the entire group. However, there are certain subgroups of $SU(2)$ that can be averaged over to recover an approximation of the Haar measure over $SU(2)$, up to the moment t . These subgroups are known as unitary t -designs of $SU(2)$. The single-qubit Clifford group, Cl_1 , is a known 2-design, and consists of only 24 elements [42].

The standard RB implementation for one sequence is presented in figure 2.4 (a), and the full sequence is as follows:

1. The qubit is initialised to $|\downarrow\rangle$.
2. A sequence of $m - 1$ single qubit Clifford gates, C_i , are applied, with each gate randomly sampled from the Clifford gate-set.

3. A final “inverting” Clifford gate, C_{inv} , is applied which is chosen such that the full Clifford gate sequence is equivalent to the identity operation.
4. A random Pauli gate, $P \in \{I, X, Y, Z\}$, is applied with equal probabilities¹.
5. The population is measured in the computational basis, with shots resulting in the expected outcome (accounting for the random Pauli) labelled as survivals, S , and opposite outcomes labelled as failures, F .
6. Steps 1-5 are repeated s times to find the average survival population, $\mathcal{P}_S$.
7. Steps 1-6 are repeated r times with different random sequences for sequence with length m .
8. Steps 1-7 are repeated for a chosen set of sequence lengths, m , here log-sampling m up to some m_{max} .
9. The average survival populations versus sequence length, m , are fit to a decay model to extract a depolarising error per Clifford gate, ε_c , and a SPAM error ε_{sp} .

The decay model used to find the fidelity versus number of Clifford gates is given by,

$$\mathcal{P}_S(m) = \frac{(1 - 2\varepsilon_{\text{sp}})(1 - 2\varepsilon_c)^m + 1}{2}, \quad (2.4)$$

where the first term in brackets is the probability of no error during SPAM, and the second, exponential term provides the probability of not depolarising during a sequence of length m .

This scheme is applied experimentally to characterise the performance of the calibrated single-qubit gates. The 24 Clifford gates are decomposed into sequences of $R_x(\pm\frac{\pi}{2})$ and $R_x(\pm\frac{\pi}{2})$, with an average of 2.17 $\frac{\pi}{2}$ -pulses per Clifford gate. Figure 2.4 (b) shows the average survival populations for a log-sampled range

¹This acts to randomise the final outcome between computational states $|\downarrow\rangle$ and $|\uparrow\rangle$, enforcing a further symmetry on any state-preparation and measurement error component.

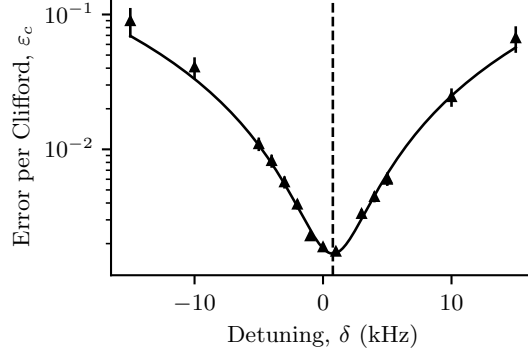


Figure 2.5: Calibrating the relative detuning between the 729-nm laser and the qubit during the application of $\frac{\pi}{2}$ -pulses. Each point corresponds to the extracted error-per-Clifford from standard RB. The solid line is the fit of a quadratic model, the dashed line is the extracted center frequency.

of sequence lengths up to $m_{\max} = 2048$ Clifford gates. Each point corresponds to an average over $r = 50$ randomisations, and $s = 50$ sample repeats per randomisation. The solid line follows equation 2.4, with fit parameters, $\varepsilon_c = 1.9(1) \times 10^{-3}$, and $\varepsilon_{\text{sp}} = 3(1) \times 10^{-3}$. The dashed line is the expected asymptote for a fully depolarised qubit. Error bars show the standard error of the average survivals for each random sequence for a given length m . From ε_c , we find a single $R_x(\frac{\pi}{2})$ average fidelity of $\mathcal{F}_{\pi/2} = 1 - \frac{\varepsilon_c}{2.17} = 99.913(2)\%$. The measured SPAM is consistent with the independently characterised state preparation (section 2.1.1) and measurement (section 2.1.2) errors of $\sim 10^{-3}$, and $\sim 10^{-4}$ respectively.

We attribute the single-qubit rotation error to the finite-temperature of the radial modes. Expanding equation ?? to second order in the Lamb-Dicke parameter, the carrier Rabi frequency is modulated depending on motional Fock state, n , as [43],

$$\Omega_n \approx \Omega \left(1 - \eta^2 \left(n + \frac{1}{2} \right) \right), \quad (2.5)$$

where η is the Lamb-Dicke parameter. For a thermal distribution each shot will result in a different rotation angle leading to a thermal dephasing error. For small rotation errors, the fidelity averaged over the Haar measure (see appendix § ??) is

given to first order by $\mathcal{F} = 1 - \frac{1}{3}\langle\theta^2\rangle$. If targetting an $R_x(\frac{\pi}{2})$ -gate, the expected fidelity due to thermal errors is then,

$$\mathcal{F}_{\pi/2}(\bar{n}) = 1 - \frac{1}{12}\pi^2\eta^4(2\bar{n}^2 + \bar{n}). \quad (2.6)$$

The radial modes for our target trapping frequencies (≈ 2.5 MHz) have Lamb-Dicke parameters of $\eta \approx 0.05$. For these experiments the radial modes are only Doppler cooled. In the following section on cooling, § 2.2.2, we will determine the Doppler limit for these modes to be $\bar{n}_d \approx 4$. Using these two estimated parameters, we expect a fidelity due to thermal errors of $\mathcal{F}_{\pi/2,\text{th}} = 99.96\%$. This is similar in magnitude to the measured value and makes the assumption cooling is optimised to reach the Doppler limit. The single-qubit gate fidelity measured here is sufficient for the projects in this thesis, however if a lower error is required, we will be required to sub-Doppler cool all radial modes.

RB can be used to calibrate systematic errors, however requiring many more samples than the RPE protocol described above. Figure 2.5 presents the detuning of the 729-nm laser with respect to the resonant frequency extracted by RPE. The points correspond to the extracted ε_c from RB sequences with the same parameters as used for figure 2.4 (b). The solid line is a quadratic fit, with center frequency $\delta = 2\pi \times 0.76(8)$ kHz. The RPE protocol does not account for the Stark-shift due to the 729-nm beam, as such we see a frequency shift between the qubit and laser when probing by RB.

We use standard RB as a daily characterisation experiment to verify consistent performance of the single-qubit gates. In chapter ?? we show further utility of single-qubit Clifford RB using a novel extension to parallel single-qubit Clifford application in a multi-qubit register.

2.1.5 Spin Coherence Times

Qubit decoherence may originate from uncontrolled fluctuations of the qubit transition frequency or from phase noise on the laser driving the qubit transition.

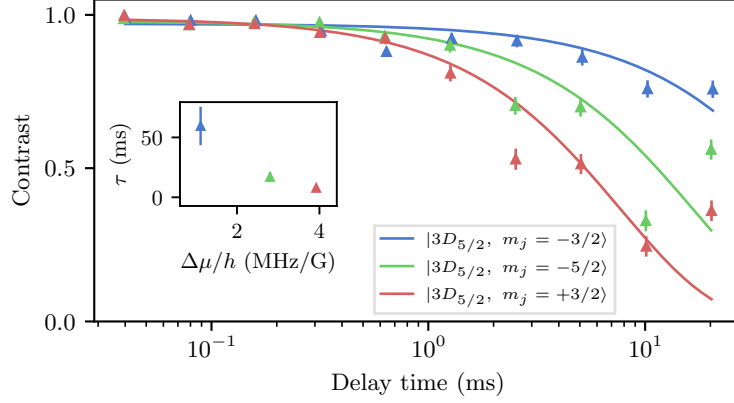


Figure 2.6: Coherence times of three quadrupole transitions with varying magnetic field sensitivities. Points correspond to the extracted contrasts of Ramsey fringes. Solid lines are exponential fits to determine the $\frac{1}{e}$ coherence time, τ . The inset plot shows the measured coherence time as a function of the magnetic field sensitivity, $\Delta\mu/h$, of the transition being probed.

Decoherence directly contributes to loss of fidelity of qubit operations, and will limit the circuit-depths possible on our apparatus.

Both electric and magnetic field noise can lead to qubit transition frequency fluctuations. The quadrupole transitions between the ground state $|4S_{1/2}, m_j = -1/2\rangle$ and the metastable manifold $|3D_{5/2}\rangle$ in $^{40}\text{Ca}^+$ are first order sensitive to any external magnetic fields. As such, even with the MuMetal magnetic shield in place, external magnetic fields are expected to be the dominant noise source. To verify this, and benchmark the current decoherence time, the Ramsey contrast is measured as a function of delay duration. The Ramsey sequence is as follows:

1. Prepare the qubit in $|\downarrow\rangle$.
2. A $R_y(\pi/2)$ -pulse is applied. preparing the state $|+\rangle = \frac{1}{\sqrt{2}}(|\downarrow\rangle + |\uparrow\rangle)$.
3. The system is left to evolve for Ramsey delay, t_{RAM} .
4. A variable $R_\phi(\pi/2)$ -pulse is applied.
5. The population of $|\downarrow\rangle$ is measured.
6. Repeat steps 1-5 for different values of ϕ to find the contrast of the Ramsey fringe.

7. Repeat steps 1-6 for different values of t_{RAM} to find the decay in contrast with time.

This Ramsey fringe contrast is directly proportional to the coherence of the resulting noisy state, ρ , given by matrix element, $\rho_{\downarrow,\uparrow}$.

Figure 2.6 shows the coherence time of three quadrupole transitions:

$|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = \{-5/2, -3/2, +3/2\}\rangle$. These transitions have magnetic field sensitivities of $\Delta\mu/h = \{2.802, 1.118, 3.921\}$ MHz G⁻¹ respectively (calculated using equation ??). The contrast decays for all three qubits are shown in the figure. The $\frac{1}{e}$ -coherence time is extracted by fitting exponential decays to the contrasts. Generally, exponential decays indicate the timescale of the noise is fast compared to the Ramsey probe durations, whilst Gaussian decays indicates comparatively slow drifts. Neither exponential nor Gaussian well describe the measured decays here due to the strong periodic structure above the exponential decay. This suggests a strong narrow-band coherent component to the noise. At the time scale of ≈ 20 ms we expect to see the effects of UK mains hum, which likely cause this structure. Regardless, fitting the exponential envelope suggests a coherence time of 17(4) ms for the qubit transition, $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$. The inset plot shows the extracted coherence time as a function of transition magnetic field sensitivity. There is a clear dependency between the coherence time and the magnetic field sensitivity, suggesting that magnetic noise is dominant.

2.1.6 Single-ion Addressing

The single-ion addressing system, introduced in § 1.6.5, is calibrated and characterised using the ion as a sensor. We verify that the beam waist at the ion plane approximately matches the designed specification, and we measure the crosstalk due to an addressed neighbouring ion. Crosstalk can cause structured coherent errors acting on the qubit register, which can have particularly adverse effects to error correction and mitigation protocols that assume independent, uncorrelated noise on each qubit. As such, designing an addressing system for low-crosstalk, and

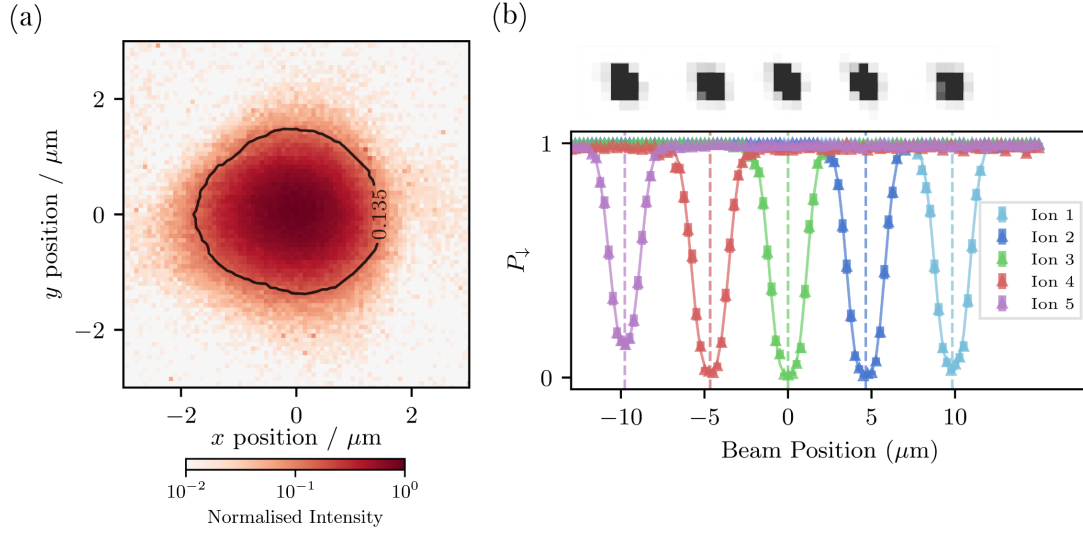


Figure 2.7: Characterisation of the single-ion addressing system using the ion as a probe. a) The normalised intensity of the focussed 729-nm beam at the ion-plane. The $1/e^2$ -waist is highlighted by the contour. b) Demonstration of spatial selective addressing by scanning the beam across a 5-ion chain while applying $R_x(\pi)$ -gates calibrated for ion 3. The probability of observing $|\downarrow\rangle$, $P_\downarrow$ is shown for each ion.

verifying its performance is critical.

The intensity profile of the beam in the ion plane is measured by Rabi oscillations on a single-trapped ion while scanning the beam position in a 2D grid. To efficiently measure this 2D array, a fixed Rabi pulse duration is used, although we note that similar measurements can be made with adaptive pulse durations, or through using RPE type protocols to improve the sensitivity of these measurement. To avoid ambiguity of the accumulated phase, we ensure that at the point of maximum field amplitude the accumulated phase does not wrap, $\theta_{\max} < \pi$. The relative optical intensity may then be extracted by considering equation 2.1, assuming a negligible decay rate, λ , and recalling that $\Omega \propto \sqrt{I}$, yielding,

$$\frac{I(x, y)}{I_{\max}} = \left(\frac{\arccos\left(\sqrt{P_\downarrow(x, y)}\right)}{\theta_{\max}} \right)^2. \quad (2.7)$$

The normalised intensity profile extracted by this method is shown in figure 2.7 (a). The qubit is initialised in $|\downarrow\rangle$ using the global addressing beam, and the total

single-addressing beam power is stabilised to 20 μW . The black contour signifies the $\frac{1}{e^2}$ waist. The spot closely resembles the expected Gaussian shape, with some aberrations, especially leading to a shoulder in the $-x$ direction. Fitting a 2D Gaussian distribution, we find $\frac{1}{e^2}$ -waists of $\omega_x = 1.584(3) \mu\text{m}$, and $\omega_y = 1.386(2) \mu\text{m}$. These closely match the target waist of $1.5 \mu\text{m}$.

The crosstalk is measured by scanning a single beam across a 5-ion chain. The axial frequency of the trap is set to 0.7 MHz, to provide an ion spacing of $\approx 5 \mu\text{m}$. When addressing the central ion of the chain, ion 3, we may observe Rabi oscillations on all qubits simultaneously to extract the relative Rabi frequency crosstalk. The Rabi frequency is determined through fitting equation 2.1, finding frequencies of $2\pi \times \{1.0(1), 1.67(5), 281.7(5), 7.96(1), 5.29(1)\}$ kHz for ions 1 to 5 respectively. This corresponds to intensity crosstalks of $\{0.12(4), 0.35(2), -, 7.99(3), 3.53(1)\} \times 10^{-4}$ respectively.

To highlight this spatial selectivity, we show the beam scanning across the 5-ion chain with a fixed pulse duration. For reference, above this plot we show fluorescence of the 5-ion chain as viewed by the imaging camera. The fixed Rabi-pulse duration is calibrated to be a $R_x(\pi)$ -pulse when acting on the central ion. Figure 2.7 (b) shows the measured populations, $P_{\downarrow}$, as a function of beam position along the axial ion-chain direction. The points correspond to populations averaged over 200 samples, with the population for each ion gathered simultaneously using the spatially resolved readout, as discussed in § 2.1.2. The solid lines follow the simple model inferred from the expected Gaussian waist of the beams,

$$P_{\downarrow,i} = 1 - \sin \left(A_i \exp \left(\frac{-(x - x_i)^2}{\omega^2} \right) \right)^2, \quad (2.8)$$

where x_i is the central position for ion, i , and A_i is included to account for a drop in field amplitude due to any misalignment between the addressing axis and the ion-chain, or due to changes in diffraction efficiency of the AODs when driving them at different frequencies. Practically we may calibrate the beam power set-point to balance any Rabi-frequencies for every addressing configuration to mitigate these

small misalignments and limited bandwidths of the AODs.

The achieved waist of $\approx 1.5 \mu\text{m}$ allows for high intensity fields at the qubits, exemplified here by the low 20 μW 729-nm beam power required for driving $\approx 1 \mu\text{s}$ $R_x(\frac{\pi}{2})$ gates. The low intensity crosstalk of $< 10^{-3}$ for neighbouring ions is sufficient for the projects demonstrated in this thesis. If lower crosstalk errors are required, it is likely this may be reduced to $< 10^{-4}$ solely through further optimisation of beam alignment through the optical train. This is suggested by the current asymmetry of crosstalk errors either side of the addressed qubit. Furthermore, as the crosstalk error is coherent, it may be further suppressed using composite pulse sequences [44].

2.2 Motion

We outline methods to track the mode frequencies, cool the motion of the qubit, quantify thermal heating of the motion, and characterise the qumode coherence time. These methods are necessary for control over qumodes for information, or for using the common motion as a means to entangle two qubits.

Motion of ions trapped in the same potential can be described by a set of normal modes [45]. These modes are labelled with respect to the ion chain geometry: axial modes oscillate parallel with the chain, and radial modes oscillate perpendicular to the chain. Due to the geometry of the 729-nm beam access with respect to the ion chain (see section 1.4), only the radial modes can be addressed in our system. For the *NPL* trap (see section 1.2), the radial modes frequencies are non-degenerate. In our system, for a single ion, the mode frequencies are approximately $\omega_{\text{upper}} = 2\pi \times 2.9 \text{ MHz}$, $\omega_{\text{lower}} = 2\pi \times 2.0 \text{ MHz}$, and $\omega_{\text{axial}} = 2\pi \times 1.0 \text{ MHz}$, where “upper” and “lower” are the two radial modes. All results in the following section are using the upper radial mode.

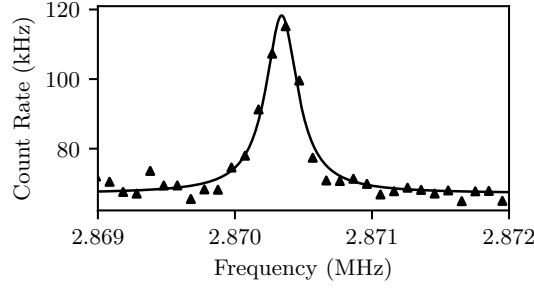


Figure 2.8: Motional mode resonance measured by RF “tickle” experiment. Points correspond to fluorescence rate measured by the camera while the solid line is a Lorentzian model fit.

2.2.1 Calibrating Mode Frequencies

The radial mode frequencies are proportional to the RF voltage (equation ??), as such any amplitude drift on the RF chain translates to mode frequency drift. The efficiency of the resonator, as discussed in § 1.2.1, is dependent on ambient temperature, and is currently not actively stabilised. As such the mode frequencies are routinely (order minutes) measured to track and update the frequency set point for interactions involving the mode.

The motional mode frequency is measured via an RF “tickle” experiment. This consists of short (≈ 1 ms) RF field pulses applied to one of the trap DC electrodes which has a non-zero electric field projection to the mode we wish to probe. The frequency of the RF field is scanned around the expected motional mode frequency, and when resonant drives the motion of the ion. Excess motional amplitude is measured by observing an increase in fluorescence when driving a closed dipole transition detuned to a non-linear region of the resonance. The motional induced sidebands at this non-linear point are unbalanced leading to a modulation of fluorescence. Here the 397-nm laser is set to provide approximately 1 saturation intensity of light to avoid power broadening the dipole transition. The 397-nm is red-detuned 100 MHz from resonance. The 866-nm is set to typical readout parameters to repump any population trapped in the D -manifold. The modulated fluorescence is integrated over ≈ 1 ms leading to an increased average count rate

when the RF tone is resonant with the mode frequency.

Figure 2.8 shows the typical fluorescence count rate for this mode frequency calibration scan on the upper radial mode. A Lorentzian profile is fit to extract central mode frequency $\omega_{\text{upper}} = 2\pi \times 2.87034(2)$ MHz, and FWHM of 280(50) Hz. The uncertainties of these parameters are determined by the bootstrapping method using 1000 re-samplings.

2.2.2 Cooling

For any interaction that involves the motion of the ion, it is essential to both prepare and measure the motional state with high fidelity. For all motional interactions considered in this thesis, the ion is initially prepared as close as possible to the motional ground state, with a mean phonon number $\bar{n} \approx 0$. This is done by an initial stage of Doppler cooling, followed by pulsed sideband cooling on the 729-nm transition. The final motional state is characterised by measuring the population in Fock states using RSB and effective BSB pulses.

Doppler Cooling

Doppler cooling relies on light incident on a moving ion appears frequency-shifted in the ion's rest frame. For Doppler cooling of $^{40}\text{Ca}^+$, both the 397-nm and 866-nm lasers are applied. The 397-nm laser is red-detuned from the dipole $|4S_{1/2}\rangle \leftrightarrow |4P_{1/2}\rangle$ transition, leading to preferential scattering of 397-nm photons that have a net cooling effect [46]. The excited $4P_{1/2}$ level decays via two channels: emission of a 397-nm photon back to $|4S_{1/2}\rangle$, or an 866-nm photon to $|3D_{3/2}\rangle$, with a branching ratio of 14.5 [47]. Since many photon scattering events are required, the cooling cycle must be closed. To prevent population trapping in the metastable $|3D_{3/2}\rangle$ level, the 866-nm laser is applied on resonance during cooling.

Equilibrium is reached when the cooling rate balances photon recoil heating. Assuming the 397-nm transition is broadened only by its natural linewidth γ ,

Parameter	Value
729-nm laser power	3.0 mW
729-nm laser detuning	-4.0 MHz*
729-nm pulse duration $N = 1$	2.5 μ s
729-nm pulse duration $N = N_{\max}$	33 μ s
Repumping 866-nm duration	14 μ s
Deshelving 854-nm duration	5 μ s
Number of cooling cycles $N_{\max}$	50

Table 2.3: *729-nm detuning is relative to the $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ transition frequency.

Experimental parameters used for sideband cooling. Other parameters for the repump and deshelve pulses are the same as for state preparation, see table 2.1. The 729-nm laser is detuned to the RSB resonance for the upper radial mode, $\omega_{\text{upper}}/2\pi = 4.0$ MHz, and the pulse duration for $N = N_{\max}$ is chosen to be the expected RSB $R(\pi)$ -pulse duration for $n = 1$.

the minimum Doppler temperature is,

$$T_{\text{Doppler}} \approx \frac{\hbar\gamma}{2k_B}, \quad (2.9)$$

where $\hbar$ is the reduced Planck constant, and k_B is Boltzmann's constant [46].

For $^{40}\text{Ca}^+$, the natural linewidth of the 397-nm transition is $\frac{\gamma}{2\pi} = 21$ MHz[48], giving $T_{\text{Doppler}} = 0.5$ mK. Given a mode frequency, the expected mean phonon number is,

$$\bar{n} = \frac{1}{e^{\hbar\omega/k_BT} - 1}. \quad (2.10)$$

For a radial mode with frequency $\omega \approx 2\pi \times 2.5$ MHz, the Doppler limit is approximately $\bar{n} \approx 4$. The parameters used here for Doppler cooling are summarised in table 2.2.

Sideband Cooling

To further cool the ions toward their motional ground state, resolved sideband cooling is used [49]. As discussed in § ??, red and blue sidebands (RSB and BSB) of the quadrupole transition are accessible, which may both flip the spin and add or remove a single phonon from the motional mode. For the $|4S_{1/2}\rangle \leftrightarrow |3D_{5/2}\rangle$ transition, at appropriate motional mode frequencies, interaction strengths, and

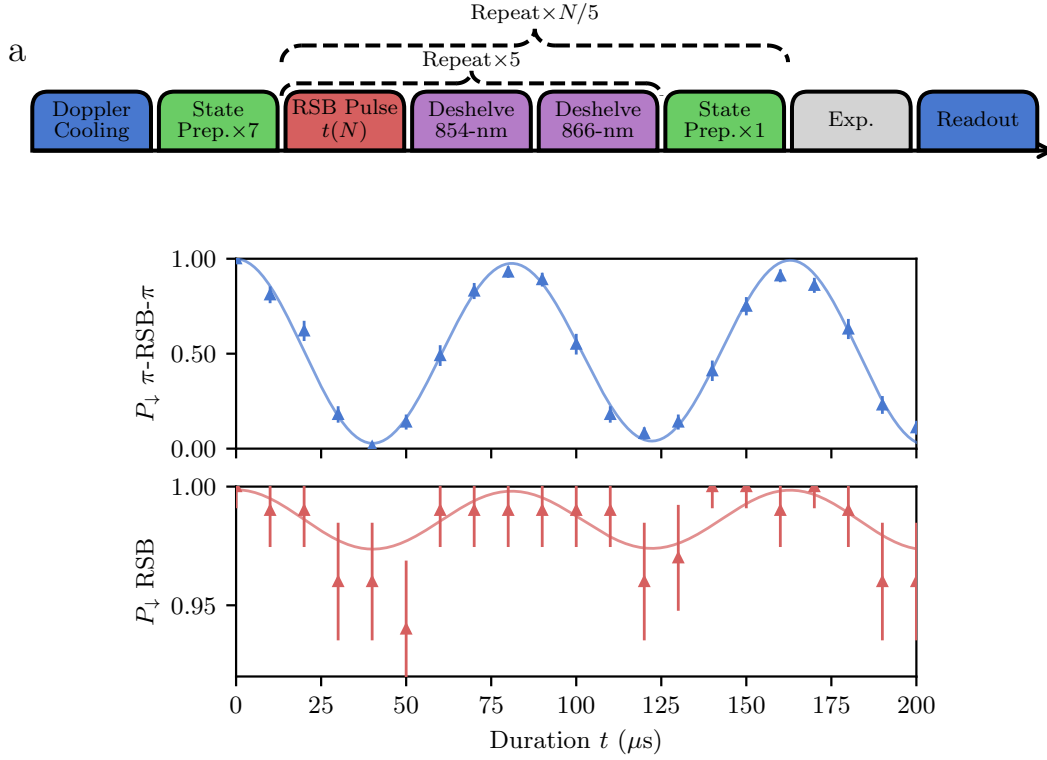


Figure 2.9: (a) Example pulse sequence for one experimental shot including pulsed sideband cooling. The *Exp.* block represents an arbitrary sequence of operations, for the experiment under study, such as a thermometry sequence. The blue blocks represent Doppler cooling and readout (section 2.1.2), the green blocks represent qubit state preparation (section 2.1.1), the red blocks are RSB pulses with varying durations $t(N)$, and the purple blocks are deshelling pulses. After every 5 sideband cooling pulses, there is an interleaved state preparation sequence to ensure there is no population trapped outside of the closed cooling cycle. The RSB length is linearly increased with pulse number N , upto the expected RSB $R(\pi)$ -pulse duration for $n = 1$. (b, c) Thermometry scans after Doppler and sideband cooling. The red points are the populations measured via on-resonance RSB pulses, and the blue points are the populations after a sequence of $R(\pi)$ -RSB- $R(\pi)$ with varying RSB probe durations. The solid lines are fits to a thermal Fock state distribution, with truncation at $n = 100$. The extracted mean occupation number is $\bar{n} = 0.03(1)$, and $\eta\Omega/(2\pi) = 11.6(1)$ kHz.

using a narrow linewidth laser (section 1.6.4), these sidebands can be resolved spectroscopically. The pulsed sideband technique employed consists of RSB pulses on the 729-nm transition, followed by deshelling, and repumping pulses on the 854-nm and 866-nm transitions respectively. For efficient cooling, initial short 729-nm pulses are applied to preferentially excite the thermal population with high phonon number, as the pulse duration for RSB $R(\pi)$ is proportional to $\sqrt{n}$, where n is the phonon number. The RSB pulse length is linearly increased to the expected RSB $R(\pi)$ -pulse duration for $n = 1$.

An example pulse sequence for a single experimental shot using sideband cooling, can be seen in figure 2.9 (a). Experimental parameters used are summarised in table 2.3.

To verify the efficacy of our sideband cooling, thermometry experiments are performed by probing Fock state populations. This is done using on resonance RSB pulses and effective BSB pulses composed of a $R_x(\pi)$ -RSB- $R_x(\pi)$ pulse sequences. The spin population dynamics are measured as RSB pulse length, t , is varied. In the case of Fock state $|0\rangle$, there should be full contrast “BSB” oscillations, and no visible oscillations when probed with RSB pulses. A thermal Fock state distribution [50] (with truncation at Fock state = 100) is fitted to these signals to extract the mean occupation number, and $\eta\Omega$, the carrier Rabi frequency multiplied by the Lamb-Dicke parameter. A typical thermometry scan after Doppler and sideband cooling can be seen in figure 2.9 (b, c), where (b) shows the population after effective BSB pulses, and (c) shows the population after RSB(t) pulses with duration t . The mean occupation number after sideband cooling is found to be $\bar{n} = 0.03(1)$, and $\eta\Omega = 11.6(1)$ kHz.

2.2.3 Heating Rates

After cooling the motion, the ion will return to a temperature in equilibrium with the environment. However, this process can be slowed by adequate isolation of the ion. In UHV ion trap systems (see section 1.4), motional heating is predominantly caused

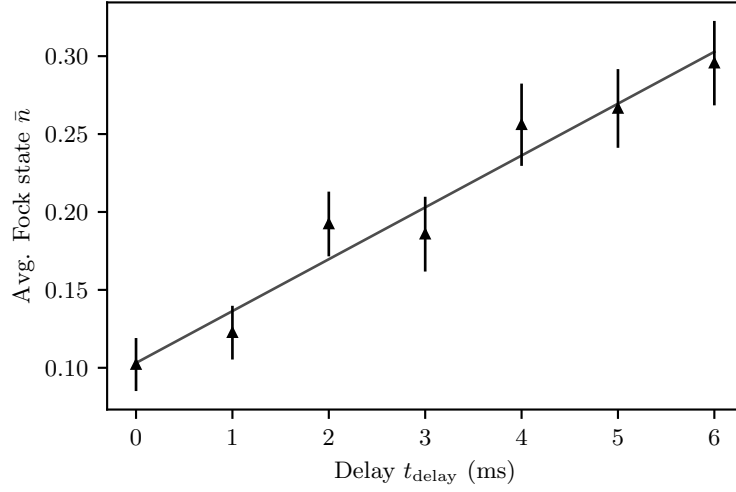


Figure 2.10: Heating rates of upper radial mode, measured by varying the delay between cooling and thermometry pulses. The solid line is a linear fit to the data, with slope $33(3)$ quanta/s. The error bars are given by the standard deviation of the fitted $\bar{n}$.

by electric field noise from the ion trap [14]. Noise due to thermal charge fluctuations at the trap surface can be mitigated by increasing ion-electrode distances. The *NPL* trap (section 1.2), used here, has an ion-electrode distance of $250\ \mu\text{m}$, larger than most surface traps, but less than that of a macroscopic blade or rod style trap [51]. Heating rates were probed by a series of thermometry experiments, explained above, with varying delay times, t_{delay} , between cooling and thermometry pulses. A typical plot can be seen in figure 2.10. The heating rate of the system was found by a linear fit, shown by the solid line, to be $33(3)$ quanta per second on the upper radial with mode frequency $\omega_{\text{upper}} \approx 3\ \text{MHz}$ on one ion. As will be discussed in section 2.2.4, the heating rate sets a limit for the motional coherence time.

2.2.4 Motional Coherence Times

For both hybrid computation, and for two-qubit entangling gates via geometric phase gates, we require motional modes with long coherence times compared to the desired circuit length. Electric field noise on the trap RF, or fluctuating external fields cause decoherence of the qumode. The field noise can cause coherent errors, motional frequency dephasing, or thermal heating of the mode, depending on its

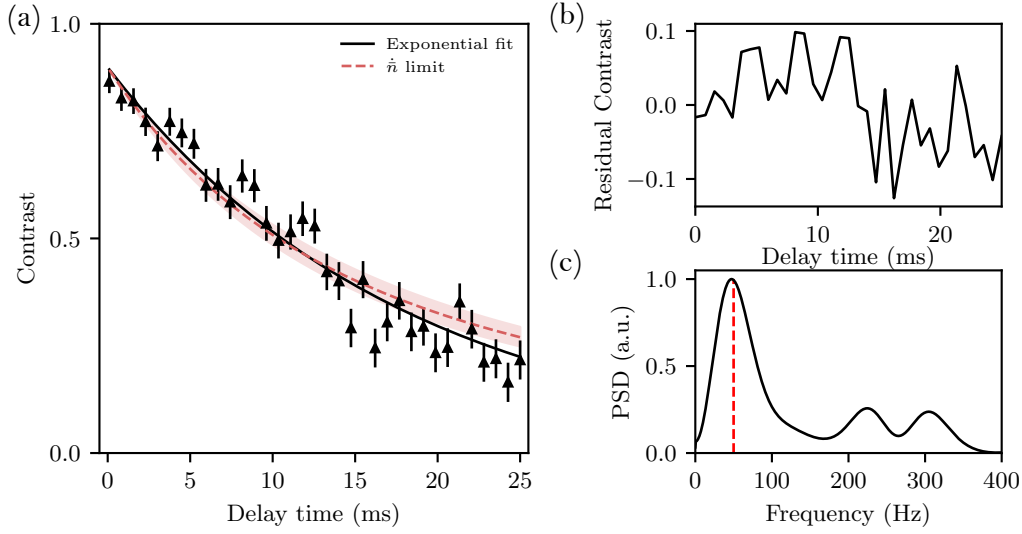


Figure 2.11: a) Motional coherence time measured by Ramsey interferometry using Fock states $|0\rangle$ and $|1\rangle$. Points are populations measured in $|\downarrow\rangle$ after the Ramsey sequence with varying delay time. The black dashed line is an exponential decay fit, with $\frac{1}{e}$ coherence time $\tau = 18.0(8)$ ms, and initial contrast $C_0 = 0.90(2)$. The red line corresponds to the expected decay due to the independently measured heating rate of $\dot{n} = 33(3)$ q/s. b) Displays the residual oscillations after subtracting the expected heating-rate profile. c) Displays the PSD of the residual oscillations, with the dashed red line signifying 50 Hz.

coherence and underlying frequency spectrum. Here the coherence time is quantified using Fock state interferometry — a Ramsey technique with initial superposition states $|\psi_{0m}\rangle = (|\downarrow, 0\rangle + |\downarrow, m\rangle)/\sqrt{2}$, where m is a pure Fock state [52]. The acquired Ramsey phase is mapped to the optical qubit through sideband pulses. Fock state interferometry is easy to implement through driving resonant carrier and RSB gates, and is sensitive to both dephasing and mode-heating noise. We note that similar interferometry techniques based on cat states [53] also quantify decoherence, however with different sensitivities to both dephasing and heating.

The Ramsey contrast decays is measured by the following sequence:

1. Prepare the ground state of both qubit and qumode, $|\downarrow, 0\rangle$. The qubit is first prepared via optical pumping, and the qumode is subsequently prepared by Doppler and resolved sideband cooling.
2. A $R_y(\pi/2)$ -pulse is applied on the carrier transition, preparing the state $(|\downarrow, 0\rangle + |\uparrow, 0\rangle)/\sqrt{2}$.

3. A RSB $R_{\text{RSB}}(\pi)$ -pulse is applied, preparing the state $|\psi_{0m}\rangle = (|\downarrow, 0\rangle + |\downarrow, 1\rangle)/\sqrt{2}$.
4. The system is left to evolve for Ramsey delay, t_{RAM} .
5. A RSB $R_{\text{RSB}}(\pi)$ -pulse is applied to map accumulated phase onto the qubit state.
6. A variable carrier $R_{\phi}(\pi/2)$ -pulse is applied.
7. The population of $|\downarrow\rangle$ is measured.
8. Repeat steps 1-7 for different values of ϕ to find the contrast of the Ramsey fringe.
9. Repeat steps 1-8 for different values of t_{RAM} to find the decay in contrast with time.

Figure 2.11 (a) shows the recovered coherence contrast for variable Ramsey delays for the upper radial mode of a single ion, with frequency $\omega_{\text{upper}} \approx 2\pi \times 2.9$ MHz. The black solid line indicates an exponential decay with a fit $\frac{1}{e}$ coherence time of $\tau = 18.0(8)$ ms, and initial contrast $C_0 = 0.90(2)$. The low initial contrast is attributed to imperfect RSB pulses due to a combination of finite residual mode temperature (expected around $\bar{n} = 0.03$), and imperfect calibration of the sideband frequency and π -gate duration. The exponential, rather than Gaussian, profile of the decay suggests that the dominating noise source is correlated for durations much shorter than the Ramsey delay times [54].

If motional-mode heating is dominant, the expected decay in the coherence between Fock states $|0\rangle$ and $|m\rangle$ for an equal superposition is given by [52, 55],

$$\rho_{0m}(t) = \frac{1}{2(1 + \dot{\bar{n}}t)^{m+1}}, \quad (2.11)$$

where ρ_{0m} is the off-diagonal density matrix element, and $\dot{\bar{n}}$ is the heating rate in phonons per second. Taking $m = 1$, and using the above protocol mapping the

motional coherence to the spin-qubit, we find spin populations,

$$P_{\downarrow}(t) = \rho_{01}(t) + \rho_{10}(t) = \frac{C_0}{(1 + \dot{n})^2}, \quad (2.12)$$

where C_0 is included to account for imperfect mapping pulses. The dashed red line overlaid on figure 2.11 (a) is the heating induced decoherence using the independently measured heating rate of $\dot{n} = 33(3)$ q/s, found in § 2.2.3. The measured contrast closely follows this decay, giving strong evidence that the coherence of the qumode is dominated by the underlying heating rate.

Further periodic structure on the decaying contrast suggests the presence of coherent noise sources. The frequencies of these oscillations are recovered by finding the power spectral density (PSD) of the residual contrast above the expected decay profile. Panel (b) displays the residual contrast amplitudes when subtracting the decay due to the heating rate from the observed populations $P_{\downarrow}$. The PSD, shown in panel (c), is found using Welch’s method with a Hann window and zero-padded to visually smooth the spectrum. The vertical dashed red line indicates 50 Hz, the electric hum from the UK AC mains supply. The PSD suggests that this hum is the dominant coherent noise source affecting the contrast decay profiles.

The effect of dephasing and heating noise depends strongly on the state of the motional mode, as such, this metric of Fock state coherence time provides evidence of adequate performance of the qumode, but does not rigourously determine the error channel acting on the oscillator. However, this measurement suggests that the coherence of the qumode is sufficient for the short hybrid sequences we plan to explore with the apparatus.

If this coherence time is not adequate for future projects, there are three possible mitigations. First, we may explore the dependence of motional mode frequency on coherence time, effectively probing the electric-field noise spectrum to find quieter regions. Second, we can decrease the impact of any incoherent noise by reducing the duration of motional interactions, however emergent off-resonant coherent errors will then require mitigation. Third, we may explore algorithmic error mitigation

techniques that reduce the impact of noise on estimated expectation values, for example, using virtual distillation. This technique was first developed explicitly to suppress the bias of an estimated observable due to finite-temperature effects [56], but has been generalised to mitigate errors occurring in quantum circuits [57, 58]. We discuss this method in the context of hybrid systems in chapter ??, and provide an initial demonstration on a spin-only system.

2.3 Spin-Motion

For full control of the spin-motion hybrid system, we require interactions that couple the two. Perhaps the simplest of this class of interactions are the red- and blue-sidebands (RSB and BSB respectively, see § ??). The spin-dependent force (SDF), as described in § ??, is another highly useful spin-motion interaction. The calibration and characterisation of the SDF is presented here. Using the SDF applied simultaneously to two qubits, we demonstrate the Molmer Sorensen two-qubit entangling gate [59].

2.3.1 Calibrating the SDF

Spin-dependent forces (SDF), as described in § ??, are used in this thesis for both two-qubit entangling gates (following section), and for generating and characterising non-linear bosonic states (§ ?? and § ??). Here we describe the calibration strategy and characterisation of a single SDF.

The SDF is generated using the bichromatic field method introduced by Molmer and Sorensen [59], using simultaneous detuned red- and blue-sidebands. In practice, this SDF interaction involves relatively few degrees of freedom to calibrate: the central qubit “carrier” frequency, the motional mode frequency, the power in each optical tone, and ensuring negligible off-resonant excitation of other spin, and spin-motion interactions.

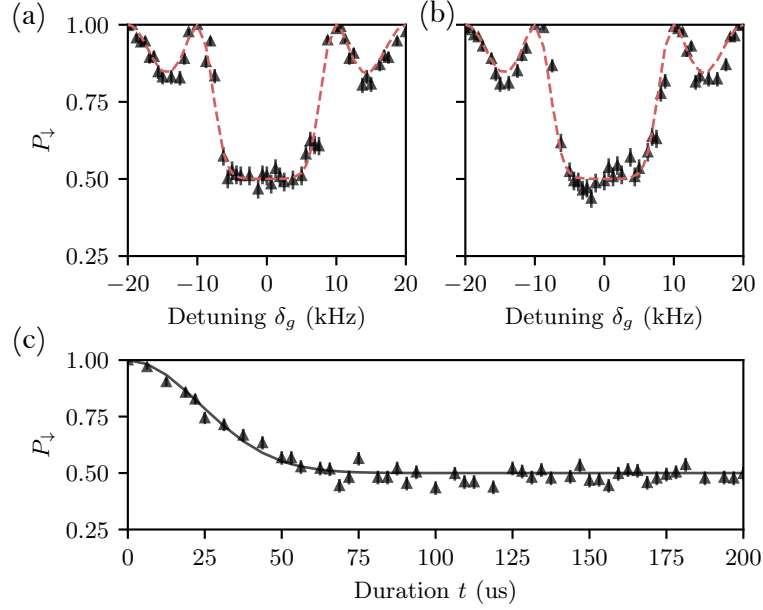


Figure 2.12: Detuning and duration scans of the spin-dependent force applied to an initial thermal state with $\bar{n} = 0.03(1)$. (a) Detuning scan, where the detuning, δ_g , is varied, and the duration is fixed at $t = 100$ μs . (b) Detuning scan with a mis-set qubit frequency of 0.5 kHz. (c) Duration scan, where the duration, t , is varied, and the detuning is fixed at $\delta_g = 0$. The solid line is a fit to the expected behaviour given by equation 2.13 with only Ω_{SDF} as a free parameter. The extracted Ω_{SDF} is used with equation 2.13 to plot the dashed red lines in (a) and (b) to guide the eye.

The bichromatic field is generated by providing two simultaneous RF tones to the single-pass (SP) AOM, as seen in the *TrapLab* panel of figure 1.10. The SDF may be driven by either the single-ion addressing system (path 729#1), or by a global field (path 729#2). The SP AOMs are driven by an arbitrary waveform generator² to provide phase, frequency, and amplitude control over the two tones. The bichromatic field is measured using a high-bandwidth photodiode after an optical pick-off before addressing the ions. The intensity measured by the photodiode has a beat-note with frequency $\omega_b \approx 2\omega_{\text{upper}}$. The contrast of this beat-note is optimised to calibrate the power balance between the two optical frequencies. The power balance is varied by updating the RF amplitude supplied to the SP AOMs. AOMs generate frequency shifts in the laser beam at the expense of small frequency-dependent angular shifts. The beam pointing of the diffracted order also has some

²Phaser Sinara board

temperature dependence. After the AOM, the beam is coupled into a single mode fibre, as such the AOMs introduce a frequency- and temperature-dependent loss. These losses can lead to significant errors in power-balance between the two tones. We found that optimising for short optical path-lengths between the SP AOM and the fibre, careful alignment to the fibre, and ensuring the SP AOM is supplied with a constant duty cycle were all required for stable power-balance.

Off-resonant excitations of other interactions are negated through selecting trap parameters to spectrally isolate a single-motional mode, and ensuring that gates are driven adiabatically — the SDF Rabi frequency, Ω , is significantly less than the frequency spacing to nearby interactions. Ensuring the interaction is smoothly ramped on and off also reduces excess off-resonant excitations.

The bichromatic field leads to a Stark shift, δ_s , of the central qubit frequency due to off-resonant coupling of both dipole and nearby quadrupole transitions. As such, the bare-qubit frequency measured by RPE in § 2.1.3 is insufficient for low-error SDFs. The Stark-shift calibration is thus done through observing the application of the SDF to the ion when all other parameters are well calibrated. When the qubit frequency is set incorrectly, by Δ , the two tones no longer have the same absolute detuning from their respective RSB or BSB. The impact of this error depends on the detuning of the SDF compared to the motional mode, δ_g . When $\delta_g \gg \Delta$ the induced phase space dynamics are similar to the desired interaction, however when $\delta_g \approx \Delta$, we see a strong deviation from ideal behaviour due to the resonant driving of only one of the sidebands. We utilise this frequency dependent non-ideal behaviour of the uncalibrated SDF to manually calibrate the induced qubit Stark shift. The phase space dynamics can be observed through projective spin measurements. An SDF square pulse of duration t_{pulse} with spin conditioning, $\hat{\sigma}_x$, and variable detuning, δ_g , is applied to the initial prepared state $|\downarrow, 0\rangle$. The spin-population returns to $P_{\downarrow} = 1$ only when the pulse duration and SDF detuning are related by $\delta_g = \frac{2\pi}{nt_{\text{pulse}}}$, for integer n . These spin dynamics are described by [60],

$$P_{\downarrow, \text{th}} = \frac{1}{2} \left(1 + e^{-4(\bar{n} + \frac{1}{2})|\beta(t, \delta_g)|^2} \right), \quad (2.13)$$

where $\bar{n}$ is the initial thermal phonon occupation, and $|\beta(t, \delta_g)| = \Omega_{\text{SDF}} \sin(\delta_g t/2)/\delta_g$, is the absolute displacement of the initial state. Deviations from this behaviour indicate either a miscalibration of Stark shift, or unintended off-resonant excitation of nearby transitions.

To give an example of this calibration procedure we show experimental data of both a well calibrated, and a miscalibrated SDF, in figure 2.12 (a), and (b) respectively. For these detuning scans we set $t_{\text{pulse}} = 100 \mu\text{s}$, and use the global addressing beam with 2 mW of total power. The points indicate the spin population averaged from 200 samples. The calibrated scan assumes a Stark-shift $\delta_s = 2\pi \times 4\text{kHz}$, while the miscalibrated scan assumes $\delta_s = 2\pi \times 3.5\text{kHz}$. The dashed red lines in both (a) and (b) are the expected population dynamics inferred using an independently measured SDF strength of Ω_{SDF} . The well-calibrated SDF shows good agreement to the expected dynamics, while the qubit-frequency mis-calibrated SDF displays a clear asymmetry near $\delta_g = 0$.

Panel (c) displays a “splitting” experiment, applying the SDF on resonance $\delta_g = 0$, and varying the duration. The solid black line is a fit of the spin populations using equation 2.13, and assuming $\bar{n} = 0$. We find here $\Omega_{\text{SDF}} = 2\pi \times 6.6(2) \text{ kHz}$. Splitting of the motional wave-packet by a resonant SDF is exactly the procedure used for motional state tomography, which we give further details for in § ??.

2.3.2 Two-Qubit Entangling Gates

To complete the demonstration of elementary operations required for universal qubit control, we characterise the performance of two-qubit entangling gates via the MS interaction on our apparatus.

This interaction, as discuss in § ??, is implemented through a detuned SDF. To reduce the sensitivity to both systematic and incoherent motional errors a Walsh modulated two-loop gate sequence is used.

The two-qubit gate performance can be characterised many different ways. Ideally a process tomography method would be used to fully characterise the CPTP

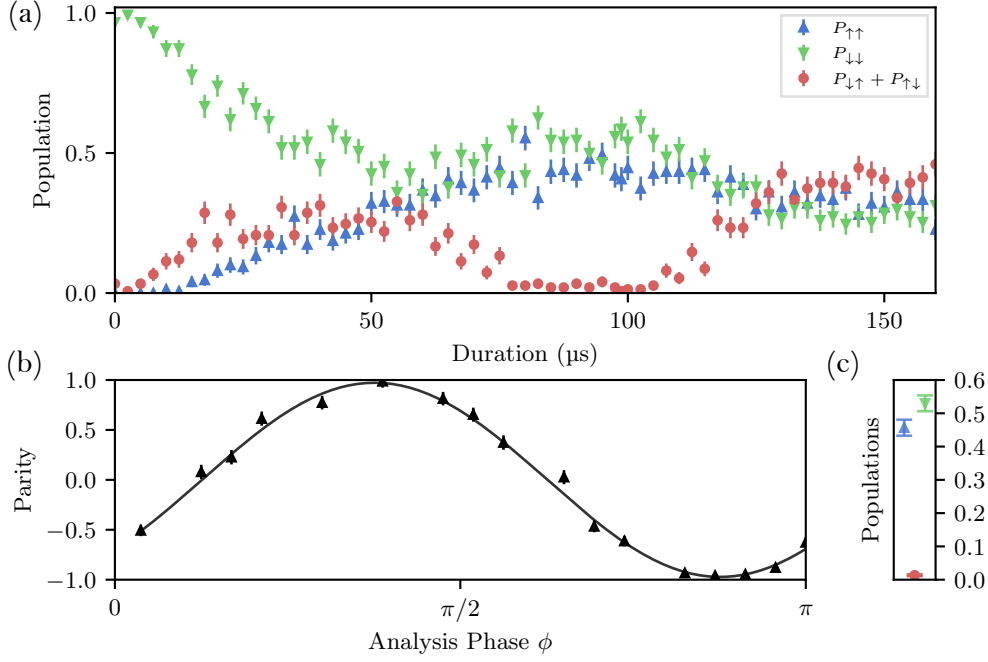


Figure 2.13: Experimental calibration and characterisation of a Walsh-modulated Mølmer-Sørensen (MS) two-qubit entangling gate with fidelity $\mathcal{F} = 0.980(6)$. (a) Populations versus duration scan of the MS interaction. Each point corresponds to the average of 200 shots, and the error bars are given by the standard deviation of these averages. The desired entangling gate corresponds here to $t_g = 95\mu\text{s}$. (b) Observed parity oscillations after the MS interaction. The fitted contrast here is $C = 0.97(1)$. (c) Populations at the desired entangling gate duration, $t_g = 95$ us.

map implemented by the noisy two-qubit gate [61]. However such a method requires near-perfect state-initialisation to correctly discern dynamics originating from the gate under test. Random circuit methods such as two-qubit Clifford RB [62], or cyclic benchmarking [63] could also be used to gain single-value performance metrics insensitive to state-preparation errors. In this thesis we use a state tomographic technique [64], which although does not rigorously characterise the action of the gate on arbitrary initial states, is quick to implement, and only requires a single additional $R_x(\frac{\pi}{2})$ -pulse.

A single $R_{\phi_s\phi_s}(\frac{\pi}{2})$ two-qubit entangling gate, where ϕ_s is some fixed angle in the xy -plane, is applied to the initial state $|\downarrow\downarrow\rangle$. This gate targets the creation of the maximally-entangled state $|\Phi_{\phi_0}^+\rangle = 1/\sqrt{2}(|\downarrow\downarrow\rangle + e^{i\phi_0}|\uparrow\uparrow\rangle)$, where ϕ_0 , is some fixed

phase³. Practically, due to systematic errors and noise on both the qubit and motional state, the final mixed state ρ is produced. The state-fidelity between ρ , and the pure Bell state is given by,

$$\mathcal{F} = \langle \Phi_{\phi_0}^+ | \rho | \Phi_{\phi_0}^+ \rangle = \frac{1}{2} (\rho_{\downarrow\downarrow,\downarrow\downarrow} + \rho_{\uparrow\uparrow,\uparrow\uparrow}) + \frac{1}{2} (e^{i\phi_0} \rho_{\uparrow\uparrow,\downarrow\downarrow} + e^{-i\phi_0} \rho_{\downarrow\downarrow,\uparrow\uparrow}), \quad (2.14)$$

where the matrix elements are given by $\rho_{ij,kl} = \langle ij | \rho | kl \rangle$. Thus to extract this fidelity, only partial state tomography is required [64]. The first bracketed term of equation 2.14, denoted A , is measured by performing projective measurements of the two ions after the applied $R_{\phi_s \phi_s}(\frac{\pi}{2})$. The extracted two-qubit joint populations directly recover this population term, $A = P_{\downarrow\downarrow} + P_{\uparrow\uparrow}$, where P_{ij} is the population of the two ions in state $|i\rangle |j\rangle$. The coherence terms in the second bracket are measured by applying a global analysis $R_\phi(\pi/2)$ pulse to the two ions with variable phase ϕ . The final spin-populations are combined to a parity measurement, defined as $\Pi = P_{\downarrow\downarrow} + P_{\uparrow\uparrow} - P_{\downarrow\uparrow} - P_{\uparrow\downarrow}$. Varying ϕ , the observed parity relates to the matrix elements of ρ as,

$$\Pi(\phi) = \rho_{\downarrow\uparrow,\uparrow\downarrow} + \rho_{\uparrow\downarrow,\downarrow\uparrow} + e^{-2i\phi} \rho_{\uparrow\uparrow,\downarrow\downarrow} + e^{+2i\phi} \rho_{\downarrow\downarrow,\uparrow\uparrow}. \quad (2.15)$$

In the case that there are no coherences between single-excitation states $|01\rangle \leftrightarrow |10\rangle$, the parity reduces to simple oscillations of the form,

$$\Pi(\phi) = 2\text{Re} [e^{-2i\phi} e^{i\phi_0} |\rho_{\uparrow\uparrow,\downarrow\downarrow}|] = C \cos(2\phi - \phi_0), \quad (2.16)$$

where we have taken $\rho_{\uparrow\uparrow,\downarrow\downarrow} := e^{i\phi_0} |\rho_{\uparrow\uparrow,\downarrow\downarrow}|$, and $C = 2|\rho_{\uparrow\uparrow,\downarrow\downarrow}|$. The contrast of the parity oscillations thus provide the magnitude of the coherence term. The Bell-state fidelity is thus given by $\mathcal{F} = \frac{A+C}{2}$.

Using this method, the fidelity of a globally addressed two-qubit entangling gate by the Walsh method is experimentally measured. Figure 2.13 (a) shows the spin-population dynamics as a function of total Walsh-1 duration for an MS gate acting on the upper radial common mode, with gate parameters, $\delta_g = -19.5$ kHz,

³ ϕ_0 is left unspecified as the target of a specific phase is not required, we only require that it is stable over time, and measured precisely.

and 35 mW of total beam power, causing a Stark shift of $\delta_s = 14$ kHz. To suppress off-resonant excitation of nearby spin-motion interactions, the SDF field is ramped on and off with a rise envelope of $\sin^2(\pi t/(2t_{\text{ramp}}))$ applied via the AWG, where $t_{\text{ramp}} = 9.6$ μs is the ramp duration. The desired geometric phase is accumulated by $t_g = 95$ μs . The parity oscillations and two-qubit populations are measured at this point, and shown in panels (b) and (c) respectively. A contrast of $C = 0.97(1)$, and population of $A = 0.987(2)$ is found, resulting in a state fidelity, $\mathcal{F} = 0.980(6)$. Each population point in these figures are found by taking the average of 200 shots of the gate sequence.

Similar fidelities were measured when applying the interaction by the single-ion addressing system on the upper radial common mode. Using gate parameters $\delta_g = -15$ kHz, $t_g = 110$ μs , and 0.148 mW of total beam power, a fidelity of $\mathcal{F} = 0.983(4)$ was measured, with contrast $C = 0.976(7)$ and population contribution $A = 0.991(1)$.

We believe we are currently limited either by systematic calibration errors, or by laser phase noise of the 729-nm system. We have anecdotally observed a strong dependence on the gate-fidelity with PID parameters for the 729-nm PDH lock. To increase the two-qubit gate fidelity, we believe that further time must be devoted to characterising the noise-spectrum of the 729-nm, and optimising lock parameters to suppress phase noise at the frequencies relevant to the two-qubit entangling gates. This, along with developing an automated gate-tuning algorithm for our apparatus, is left to future work. The current fidelities observed here are sufficient for the projects outlined in this thesis. In § ?? we calibrate $R_{xx}(\frac{\pi}{4})$ gates using this state-fidelity approach.

2.4 Summary

These results serve as both a proof of principle for full discrete-variable control on our system, but also as a benchmark to which we can compare future system

Operation	Fidelity	Section
State Prep	99.96(2)	§ 2.1.1
Measurement	< 99.7(3)	§ 2.1.4
$R_x(\pi/2)$	99.913(2)	§ 2.1.4
$R_{xx}(\pi/2)$	98.3(4)	§ 2.3.2

Table 2.4: Summary of qubit-operation fidelities presented in this chapter.

improvements. The fidelities of the operations demonstrated in this chapter are summarised in table 2.4, with relevant sections highlighted.

Appendices

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